Experiments done

Color Temperature Experiments

For the Color Thermochromic pigments (where the range of activation temperatures is 25°C-31°C for Yellow, Red, Blue), take pictures at each degree from 24°C-32°C in order to see how the color changes.
In order to do so, use a 3D printing bed to choose the temperature, and then use an infrared non-contact thermometer to check whether the temperature of the bed has reached the intended temperature.

First Attempt - 01/04/26

Full Color DragonSkin20 (20g DragonSkin, 0.16g Yellow, 0.10g Red, 0.10g Blue)

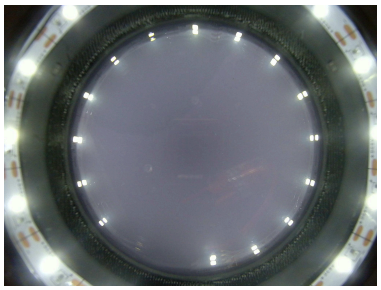
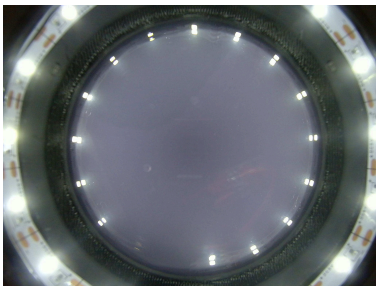
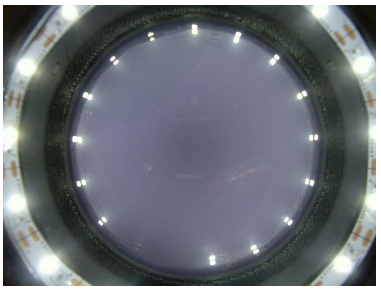
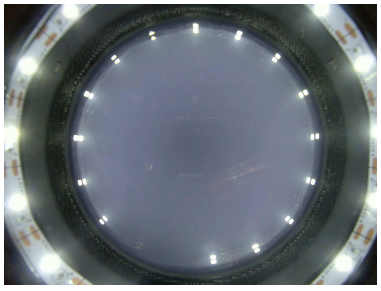
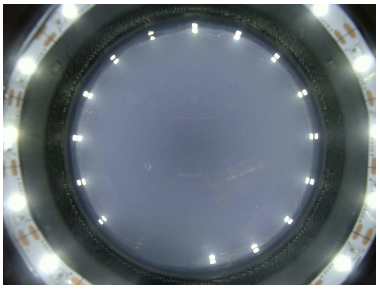
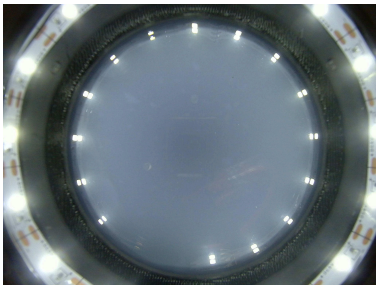
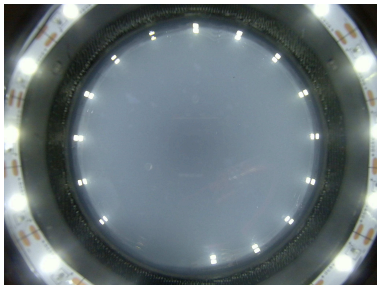
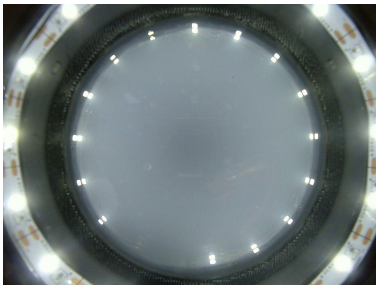
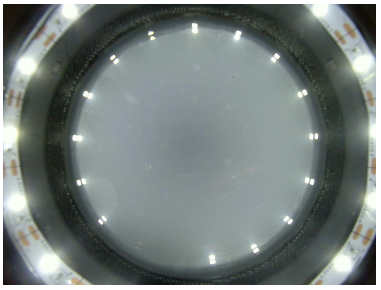
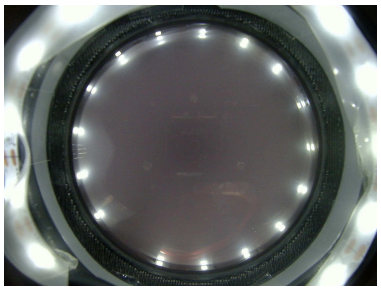
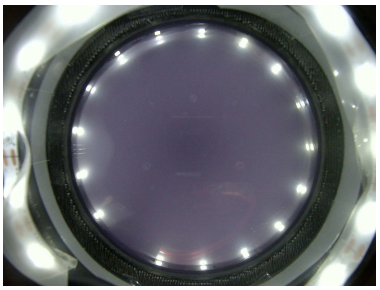
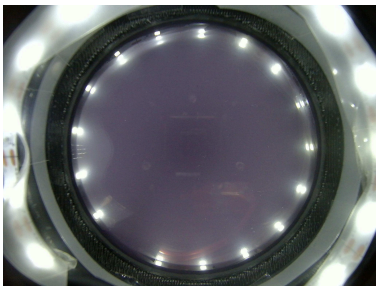
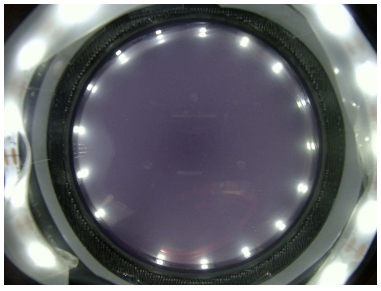
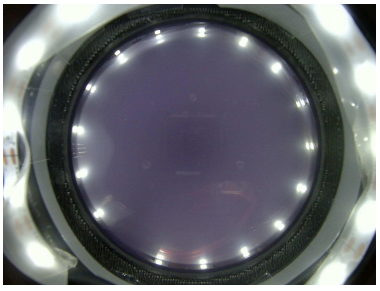
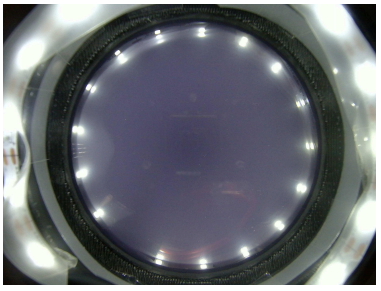
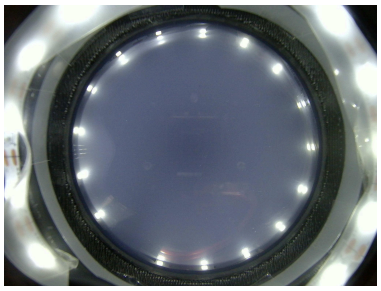
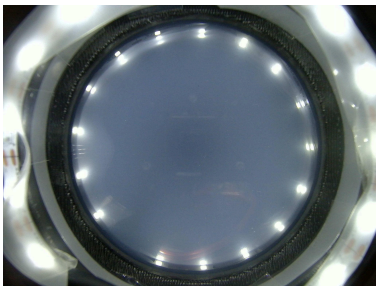
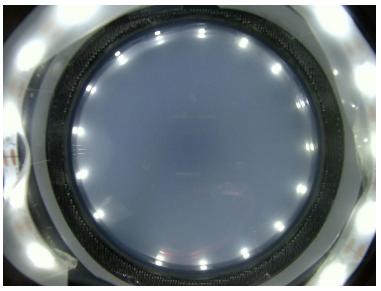
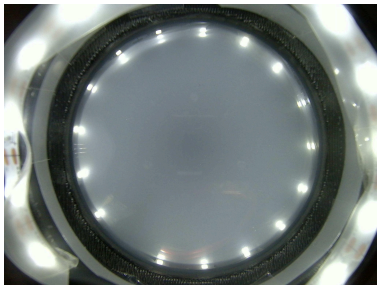
 24°C	 25°C	 26°C
 27°C	 28°C	 29°C
 30°C	 31°C	 32°C

Table showing the Color TLCs at each degree between 24°C-32°C for the ratio mentioned above

Second Attempt - 02/04/26

Full Color DragonSkin20 (22g DragonSkin, 0.5g Yellow, 0.3g Red, 0.3g Blue) - thickness is 1mm with the adjustable film applicator → measured 0.49 on one side, 0.39 on the other

Still need to fix the dark spot in the middle → in this one, there is diffuse paper that covers the LED ring and well as diffuse paper on the outer ring

 Room temperature	 24°C	 24°C
 25°C	 26°C	 27°C
 28°C	 29°C	 30°C
 31°C	32°C	

Full Color Experiment - 28/04/26

Silicone Fabrication

DragonSkin20 - Add part A first, then the thermochromic pigments (Yellow, Red, Blue), mix, add some white dye, mix, then add part B (at 1:1 ratio with part A), mix, add some more white dye, mix, then add silicone solvent, mix

DragonSkin20 fabrication - 02/04/26
22g DragonSkin, 0.5g Yellow pigment, 0.3g Red Pigment, 0.3g Blue Pigment, 2g Silicone solvent, 20 drops of white dye

DragonSkin20 fabrication - 14/04/26
20g DragonSkin, 0.5g Yellow pigment, 0.3g Red Pigment, 0.3g Blue Pigment, 2g Silicone solvent, 20 drops of white dye, thickness is 1mm on adjustable film applicator 8g DragonSkin, 0.12 pigment for temperatures (10°C, 18°C, 38°C, 45°C, 70°C), 1g of silicone solvent, thickness is 0.2mm on adjustable film applicator

DragonSkin20 fabrication - 17/04/26
20g DragonSkin, 0.5g Yellow pigment, 0.3g Red Pigment, 0.3g Blue Pigment, 2g Silicone solvent, 20 drops of white dye, thickness is 1mm on adjustable film applicator 8g DragonSkin, 0.12 pigment for temperatures (10°C, 18°C, 38°C, 45°C, 70°C), 1.1g of silicone solvent, thickness is 0.1mm on adjustable film applicator

Solaris fabrication - 21/04/26
45g Solaris, 1.5g curing agent

PsychoPaint fabrication - 23/04/26
12g PsychoPaint, thickness is 0.2mm on adjustable film applicator, 0.18g pigment for temperatures (10°C, 18°C, 38°C, 45°C, 70°C), 0.22g pigment for temperatures (25°C, 28°C, 31°C), 1.2g of silicone solvent

PsychoPaint fabrication - 23/04/26
12g PsychoPaint, thickness is 0.35mm on adjustable film applicator, 0.18g pigment for temperatures (10°C, 18°C, 38°C, 45°C, 70°C), 0.22g pigment for temperatures (25°C, 28°C, 31°C), 4g of silicone solvent

PsychoPaint fabrication for grating (trial with no dye or thermo pigments) - 23/04/26
6g PsychoPaint, 12g of Novocs Matte, used 3D printed grating molds (1mm and 2mm periods with 0.2mm and 0.4mm thickness)

PsychoPaint fabrication - 30/04/26

16g Psychopaint, 3.2g of Novocs Matte, 0.25g pigment for temperatures (10°C, 18°C, 38°C, 45°C, 70°C), 0.3g pigment for temperatures (25°C, 28°C, 31°C) → will put color on top of this for experiments

Steps to take

Colored layer

Materials required: DragonSkin20 silicone, Novocs Matte (silicone solvent), white dye, adjustable film applicator, Thermochromic Pigments (Yellow with activation temperature of 25°C, Red with activation temperature of 28°C, Blue with activation temperature of 31°C), transparent sheet of paper with at least one smooth side, disposable plastic cup, small disposable wooden spoon, Scale (to measure weight), IPA

Access required to a vacuum chamber in order to remove bubbles from silicone mixture, as well as a flat surface in order to make colored layer from film applicator

Ratio chosen as of 17/04/26: Total 20g of DragonSkin20 (10g Part A and 10g Part B in 1:1 ratio), 2g of Novocs Matte, 20 drops of white dye, 0.5g of yellow pigment, 0.3g of red & blue pigment

Adjustable film applicator set to 1mm in order to obtain 500µm thickness

Pot life of DragonSkin20 is 25 minutes, once Part B is added, you will only have 25 minutes to complete the remaining steps

1. Place plastic cup on scale (you can reset each new measurement to zero in order to make it easier) and then add 10g of DragonSkin20 Part A
2. Add 10 drops of white dye and then mix with wooden spoon (spoon can be left in cup and be used for all remaining mixing)
3. Add 0.5g of yellow pigment, mix, then 0.3g of red pigment, mix, then finally add 0.3g of blue pigment, mix
4. Add 10g of DragonSkin20 Part B and then mix (25 minute countdown starts now)
5. Add remaining 10 drops of white dye and then mix some more
6. Add 2g of Novocs Matte (1:10 ratio) and then mix
7. Put cup inside vacuum chamber (have the cup in a plastic container to prevent it falling over in the chamber) and then remove all bubbles from silicone mixture
    a. Silicone mixture may start to boil at which case once you return to atmospheric pressure, the bubbles will have gone
8. Pour mixture onto smooth side of the transparent sheet (which should be laid on a flat surface) and then use film applicator (which is set to 1mm on both sides) to smooth the mixture over the length of the sheet
9. Clean film applicator with IPA
10. Leave the silicone mixture to cure for 4 hours

Final result should look something like the picture on the left

Black layer

Materials required: Same as for colored layer, Black Thermochromic Pigments (Activation temperatures of 10°C, 18°C, 25°C, 28°C, 31°C, 38°C, 45°C, 70°C), Psycho Paint Silicone

The black layer will be added on top of the colored layer, therefore the steps to the colored layer will have to be done first and be fully cured before attempting to make the black layer

Pot life of Psycho Paint is 40 minutes, once Part B is added, you will only have 40 minutes to complete the remaining steps

Ratio chosen as of 17/04/26: Yet to be decided (8g?)

Adjustable film applicator set to ? (yet to be decided) (do not recalibrate between doing the colored layer and the black layer!)

In order to try and make black layer thin, try and make the ratio of solvent to silicone 1:3

1. Repeat the exact same steps for the colored layer and only start step 2 once it is fully cured
2. Place new plastic cup on scale and then pour 4g of Psycho Paint Part A
3. Add each pigment at the required weight (see below) and then mix after having added each new pigment with new wooden spoon
    a. 0.12g for black pigments with activation temperatures of 10°C, 18°C, 38°C, 45°C, 70°C
    b. 0.15g for black pigments with activation temperatures of 25°C, 28°C, 31°C
4. Add 4g of Psycho Paint Part B (40 minute countdown starts now)
5. Add 0.8g of Novocs Matte (1:10 ratio) and then mix
6. Put cup inside vacuum chamber (have the cup in a plastic container to prevent it falling over in the chamber) and then remove all bubbles from silicone mixture
    a. Silicone mixture may start to boil at which case once you return to atmospheric pressure, the bubbles will have gone
7. Pour mixture on top of the colored layer (still on a flat surface) and then use film applicator (which is set to 1.2mm on both sides) to smooth the mixture over the length of the sheet
8. Clean film applicator with IPA
9. Leave the silicone mixture to cure for 2.5 hours

Grating

Materials required: DragonSkin20 silicone, Psycho Paint silicone, Novocs Matte (silicone solvent), white & black dye, adjustable film applicator, transparent sheet of paper with at least one smooth side, disposable plastic cup, small disposable wooden spoon, Scale (to measure weight), Screen Printing Frame, IPA, Air Pump

The grating is done in two steps, first by making the base layer white with DragonSkin20, and then by screen printing the the black grating onto the white layer

The quantities in the steps below are for a single sheet (you can scale it up if you want to make more)

Part 1

1. Place plastic cup on scale and then pour 10g of DragonSkin20 Part A
2. Add 10 drops of white dye and then mix
3. Pour 10g of DragonSkin20 Part B and then mix (25 minute cooldown starts now)
4. Add 10 more drops of white dye then mix
5. Add 2g of Novocs Matte (1:10 ratio) then mix
    a. If the silicone mixture does not appear fully white when looking through the cup, add more white dye until it looks fully white
6. Put cup inside vacuum chamber (have the cup in a plastic container to prevent it falling over in the chamber) and then remove all bubbles from silicone mixture
    a. Silicone mixture may start to boil at which case once you return to atmospheric pressure, the bubbles will have gone
7. Pour mixture onto smooth side of the transparent sheet (which is on a flat surface) and then use film applicator (which is set to 1mm on both sides) to smooth the mixture over the length of the sheet
8. Clean film applicator with IPA
9. Leave the silicone mixture to cure for 4 hours

Part 2 - Screen Printing with frame

1. Using a new plastic cup, pour 2.5g of Psycho Paint Part A
2. Pour 2.5g of Psycho Paint Part B (45 minute cooldown from now)
3. Add black dye until mixture is very dark
4. Add Novocs Matte to the mixture at a 2:1 ratio, so 10g in this case
5. Place white silicone beneath screen printing frame
6. Pour some of the black ink into frame on the end and then pass through it gently first with rectangular press, and then go back while pressing hard so that you make contact with the white silicone this time
    a. The screen printing can only be done in one motion back and forth, after that do not try and redo
7. Clean the screen printing frame with a brush, the Novocs Matte (silicone solvent), and some IPA, make sure that the transparent bits of the screen printing frame (the middle section) are most clean

8. After that, use the air pump to clean the transparent bits on both sides until it is fully clean

Part 2 - Screen Printing with 3D printed masks

1. Apply double sided tape onto the smooth edges of the mask
2. Apply the silicone mold spray
3. Stick onto film paper
4. Using a new plastic cup, pour DragonSkin20 part A
5. Pour an equal amount of DragonSkin20 part B
6. Add 15 drops of black dye
7. Put cup inside vacuum chamber (have the cup in a plastic container to prevent it falling over in the chamber) and then remove all bubbles from silicone mixture
    a. Silicone mixture may start to boil at which case once you return to atmospheric pressure, the bubbles will have gone
8. Pour some black ink onto the edge of the mask and then press through it with rectangular press, try to only do it in one time, but if doing it more than once then make sure to continuously do it in the same direction
9. Leave to cure

Elastomer

Materials required: Elastomer (Smooth On Solaris), Curing Agent, Scale, disposable plastic cup, small disposable wooden spoon, IPA, Smooth On Primer

Access required to a vacuum chamber in order to remove bubbles from silicone mixture, and a glass to cover the prototype, the top part+glass should already be 3D printed, laser cut, and hot glued together

1. Place the plastic cup on the scale and then pour 1.5g of curing agent
2. Subsequently add 45g of the elastomer
3. Use the wooden spoon to mix the two fully
4. Put into the vacuum (turn on button with blue handle parallel and red handle at right angle and make sure to press down on lid) and remove all the bubbles (this will take about 20 minutes if the surface area is small)
5. Take the top part setup and use IPA to clean the glass
6. Put primer onto glass, pump is broken so just use tail
7. Pour the mix onto the setup until it fills all the space (make sure to be consistent with pour in order to avoid bubbles, so don’t stop and start)

Glue Silicone to Glass

Materials required: Solaris, Silicone purer, IPA

Already done: 3mm acrylic glass of dimension 81.5mmx81.5mm should be already laser cut, Top part 3D printed, DragonSkin20 should already be made

1. Hot Glue glass to 3D printed top part
2. Use IPA to clean glass and the purer with tail
3. Make 10g of Solaris (5g Part A and 5g Part B), and no silicone solvent
4. Put Solaris into vacuum and remove all bubbles
5. Cut the DragonSkin into a piece that is slightly bigger than needed (including the film paper itself)
6. Without touching the clean side (side that remained stuck to the film paper), flip it over
7. Pour Solaris onto one side and then very slowly, and at an angle, press down on it with the glass in order to help press away the bubbles
8. If it needs to be moved, please move very slowly across

3D printed grating

1. Apply solution on just the grating, not the film
2. Use double sided tape on just the smooth side

Force-Depth Experiment

1. Put the circular indenter tip onto the MARK10 machine
2. Set the max force to 50N
3. Start Experiment
4. Save Data
5. Repeat 5 times
6. Export Data via Google Drive
7. Only take the depth at every 1N step via an average of values at that given force

Force-Stripe Width Experiment

1. Put the circular indenter tip onto the MARK10 machine
2. Set the max force to given force value (steps of 3N until 48N)
3. Take picture for reference
4. Start Experiment
5. Take picture of deformed image
6. Repeat 5 times
7. Repeat for all steps

Then knowing where the deformation is, it is possible to then calculate the frequency of the stripes and how that changed compared to when it was undeformed

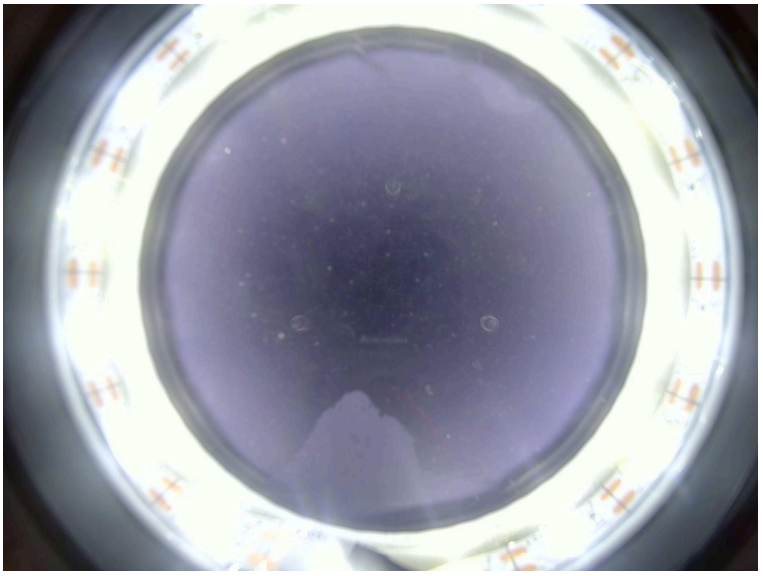
Different setup attempts

This tab seeks to document some of the attempts to achieve the best possibility for the lighting when doing the temperature experiments. The goals I am looking for in order to obtain the best lighting are the following:

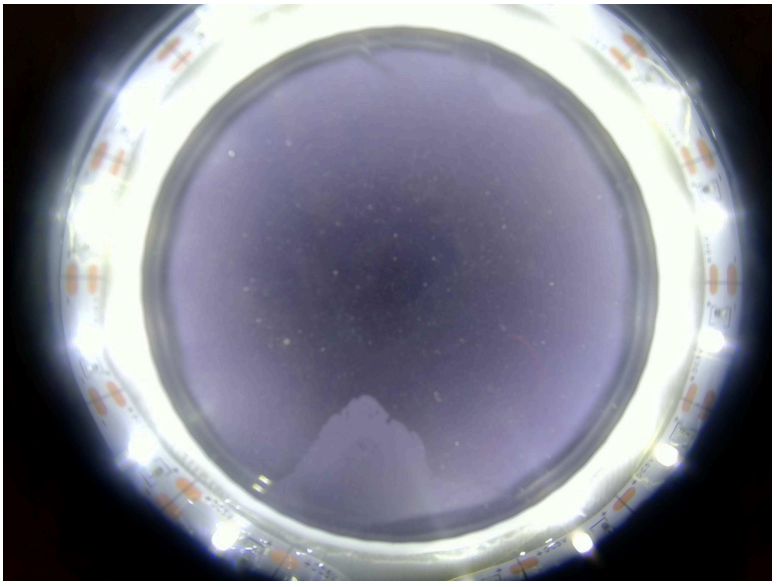
- Good brightness (image is clear, not too dark/bright)
- Uniform illumination
- No reflection
- Maximize region of interest

Here is a list of the different permutations with the respective image:

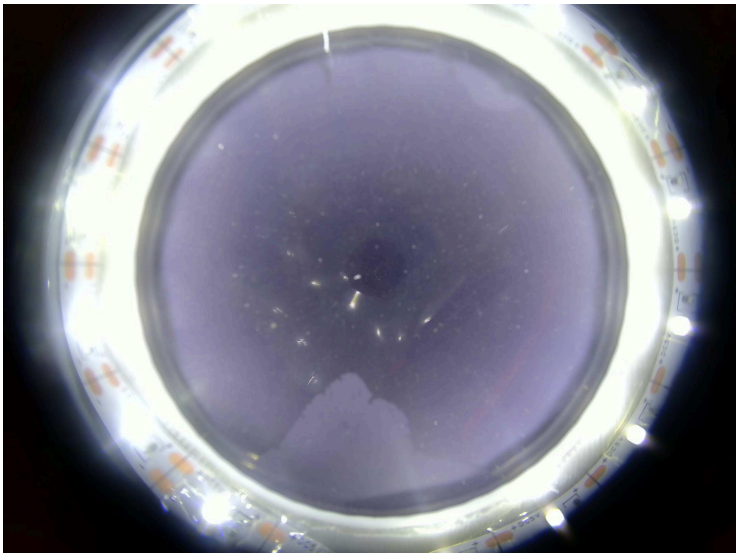
White Top, Black Core, LED strip as close as top as possible



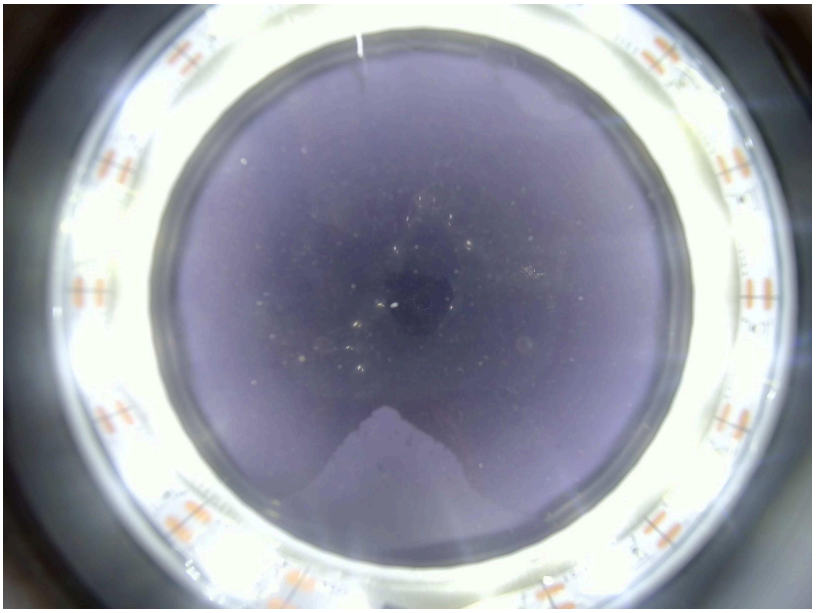
White Top, Black Core, LED strip as close as top as possible, anti-reflective shield



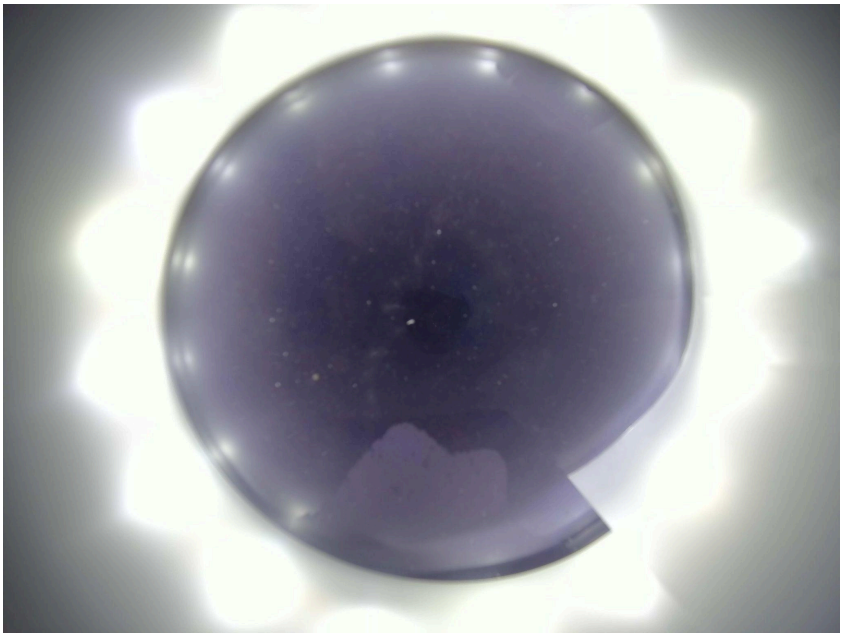
White Top, Black Core, LED strip as close as top as possible, anti-reflective shield with diffuse paper on shield



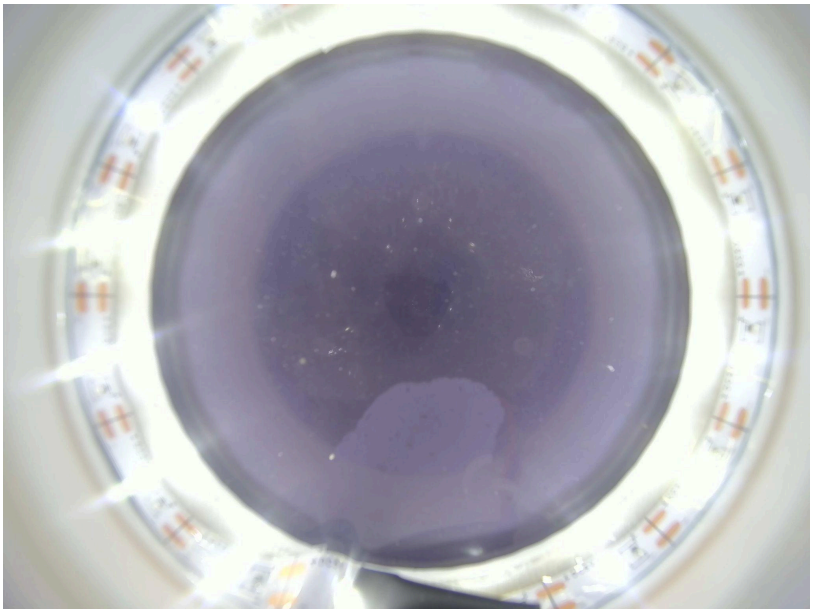
White Top, Black Core, LED strip as close as top as possible, diffuse paper as shield



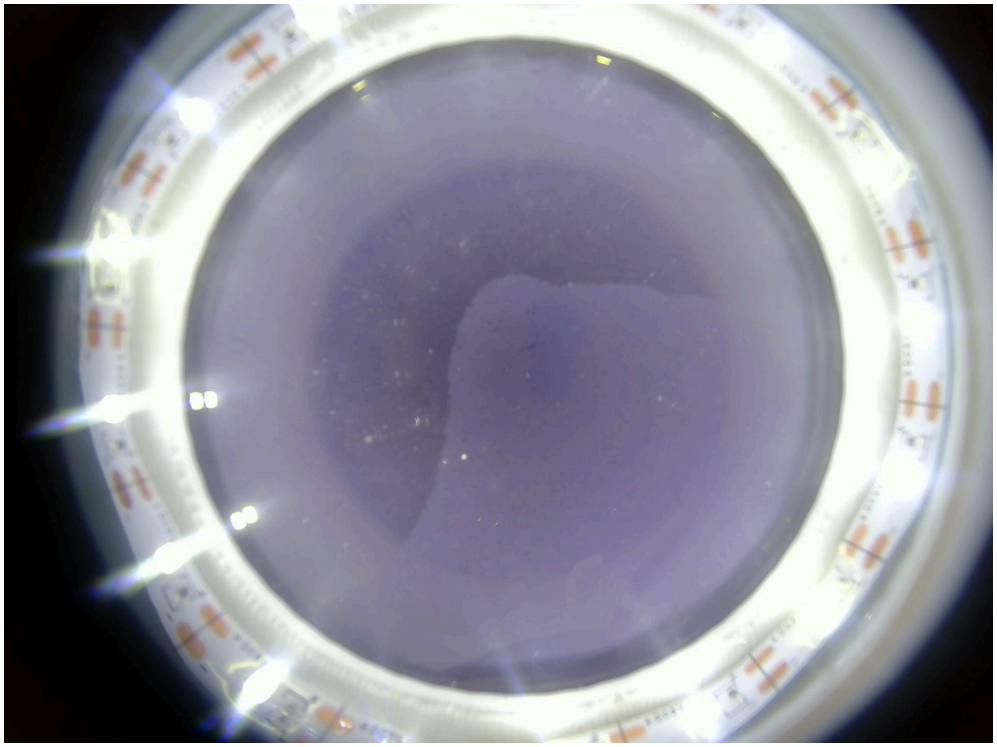
White Top, Black Core, LED strip as close as top as possible, diffuse paper as shield, diffuse paper lining the interior at a few millimeters from light source



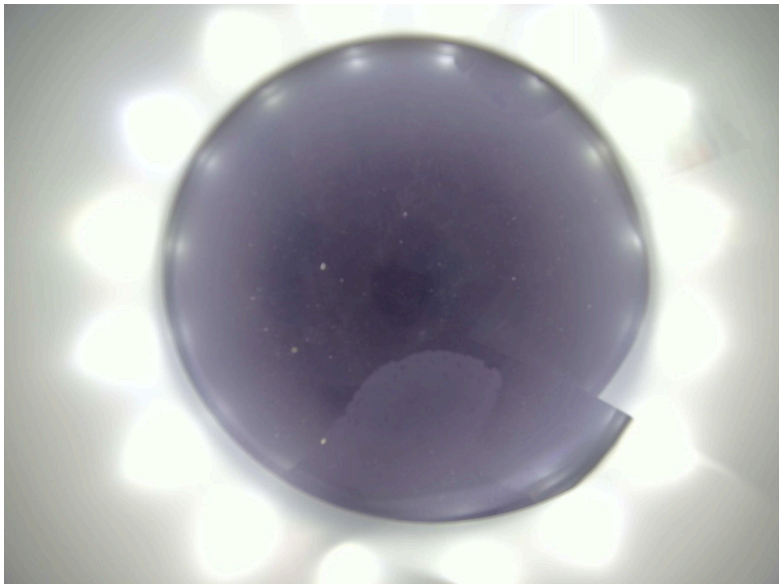
White Top, White Core, LED strip as close as top as possible, diffuse paper as shield



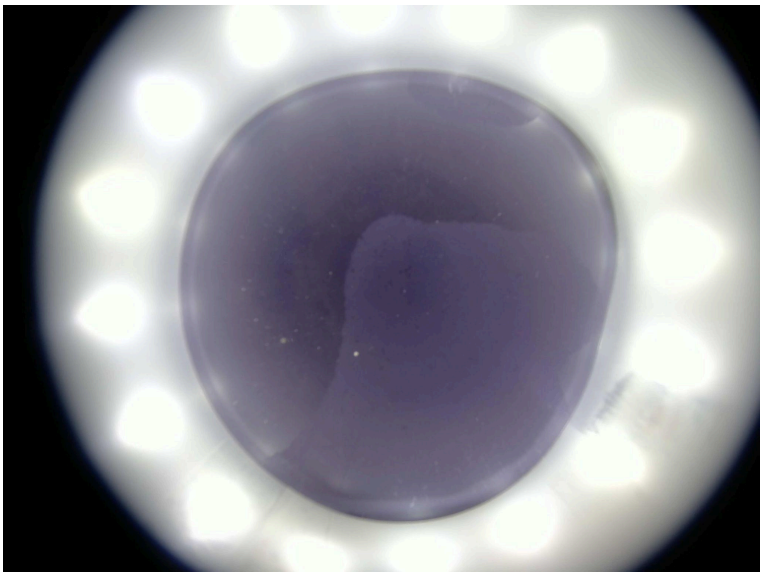
White Top, White Core, LED strip as close as top as possible, diffuse paper on top of anti-reflective shield



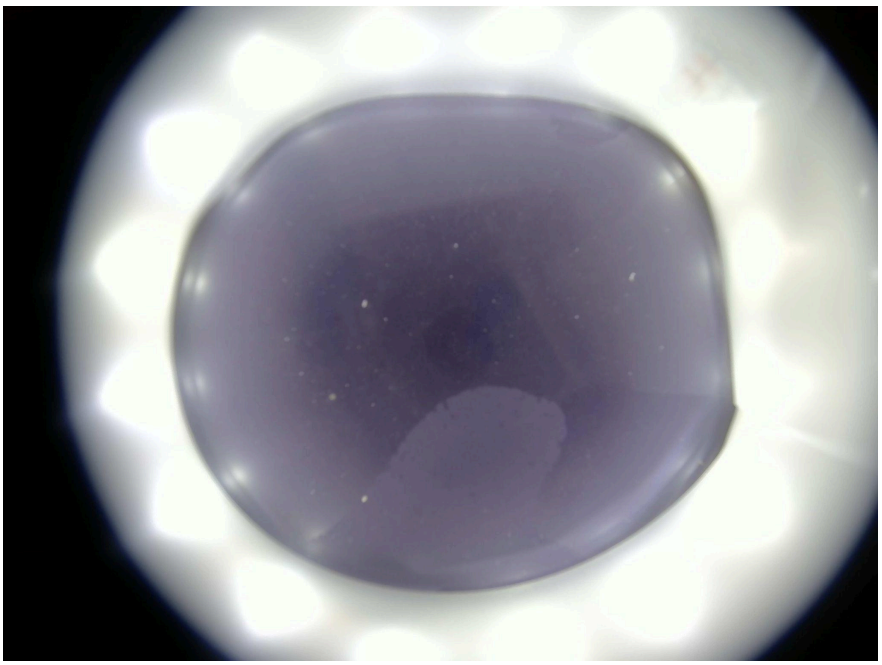
White Top, White Core, LED strip as close as top as possible, diffuse paper as shield, diffuse paper lining the interior at a few millimeters from light source



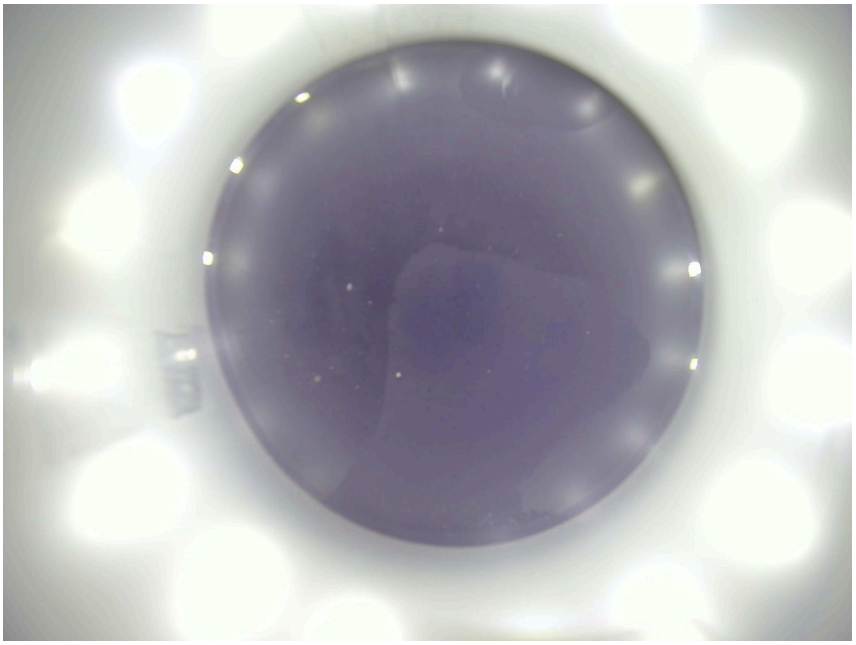
White Top, White Core, LED strip as close as top as possible, diffuse paper on top of anti reflective shield, diffuse paper lining the interior at a few millimeters from light source



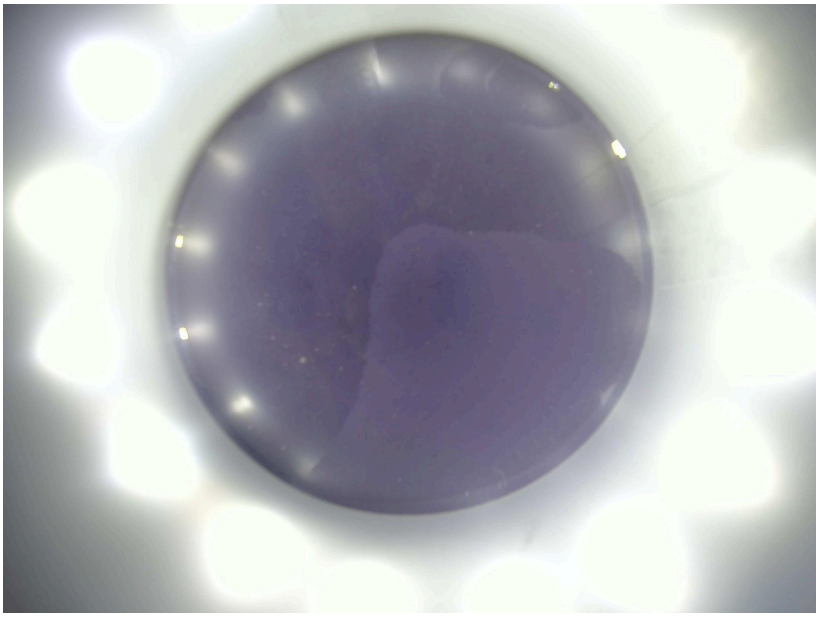
White Top, White Core, LED strip a bit further down, diffuse paper on top of anti reflective shield, diffuse paper lining the interior at a few millimeters from light source



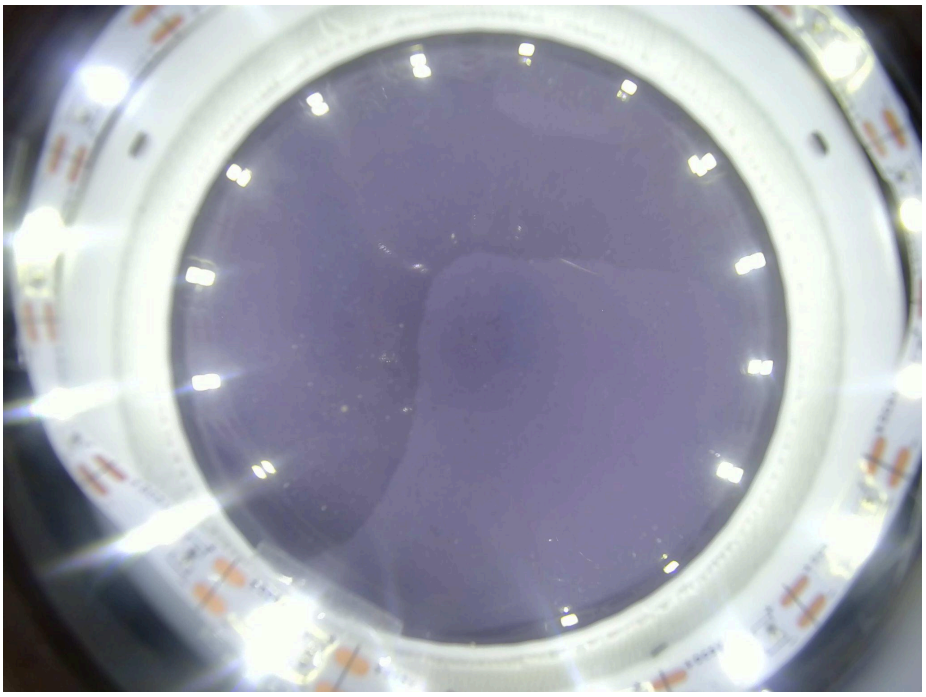
White Top, White Core, LED strip a bit further down, diffuse paper as shield, diffuse paper lining the interior at a few millimeters from light source



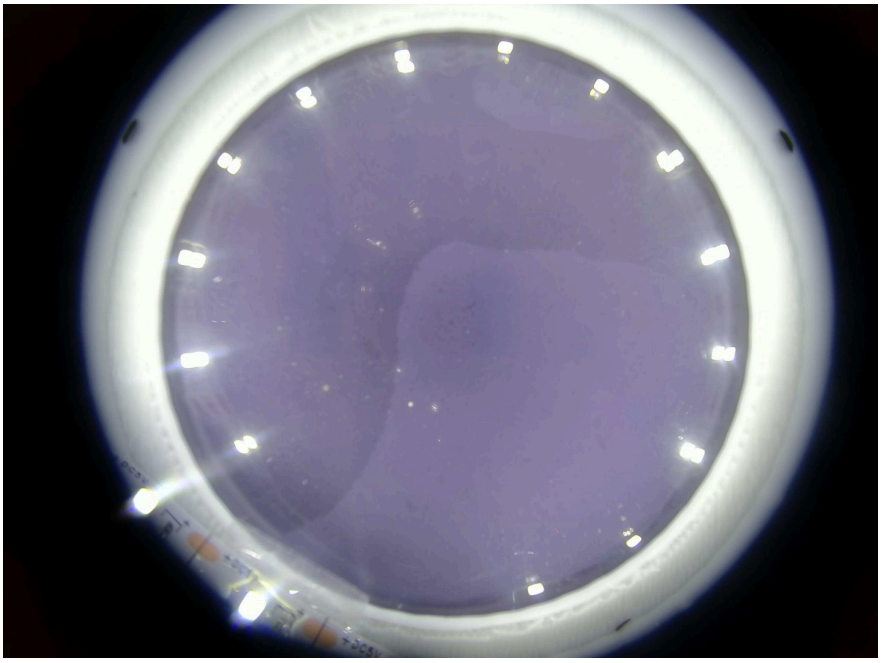
White Top, Black Core, LED strip a bit further down, diffuse paper as shield, diffuse paper lining the interior at a few millimeters from light source



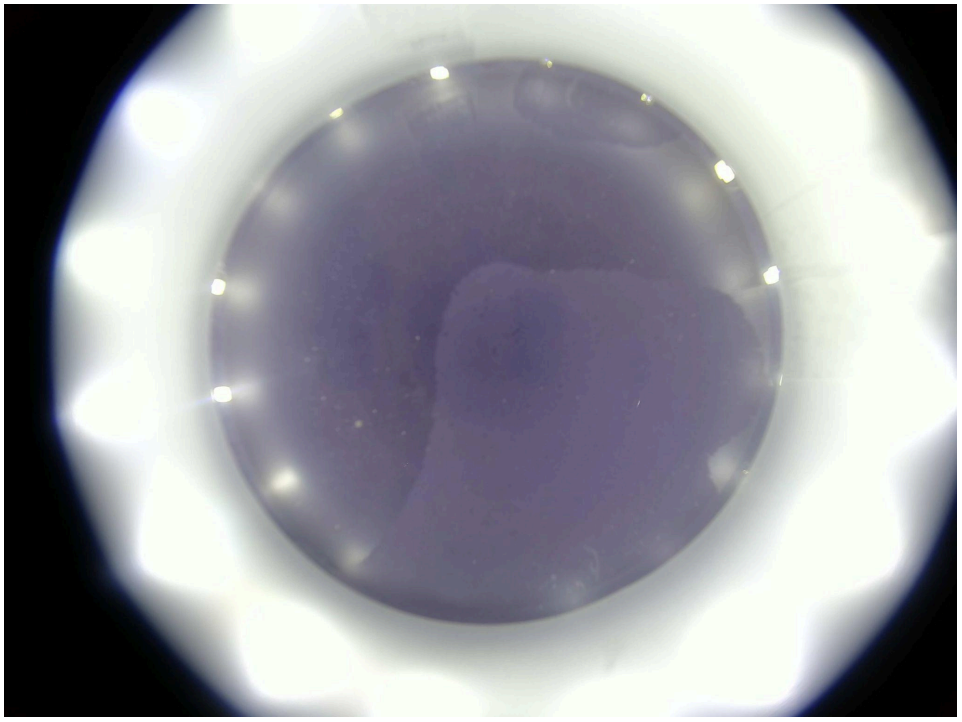
White Top, Black Core, LED strip a bit further down, diffuse paper as shield



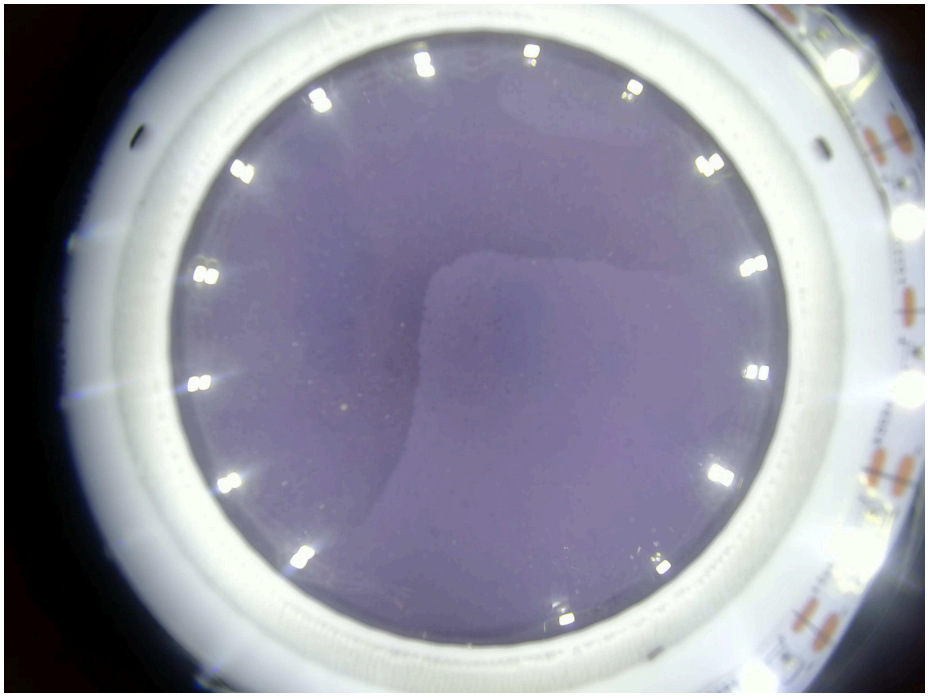
White Top, Black Core, LED strip a bit further down, anti-reflective shield with diffuse paper on shield



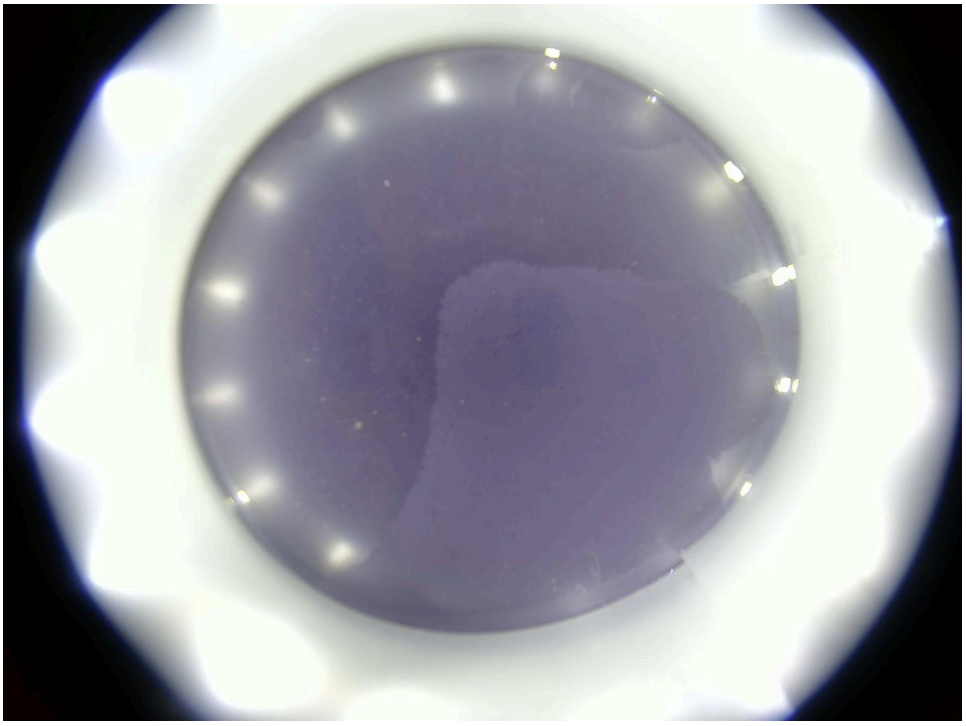
White Top, Black Core, LED strip a bit further down, diffuse paper on anti-reflective shield, diffuse paper lining the interior at a few millimeters from light source



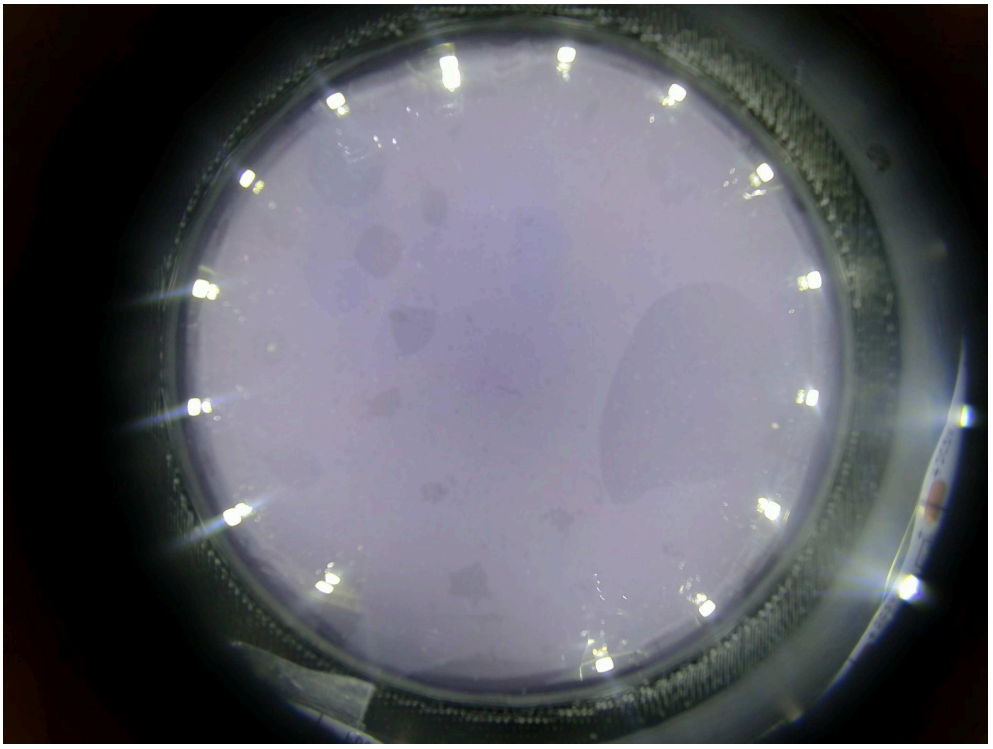
White Top, Black Core, LED strip a bit further down, anti-reflective shield



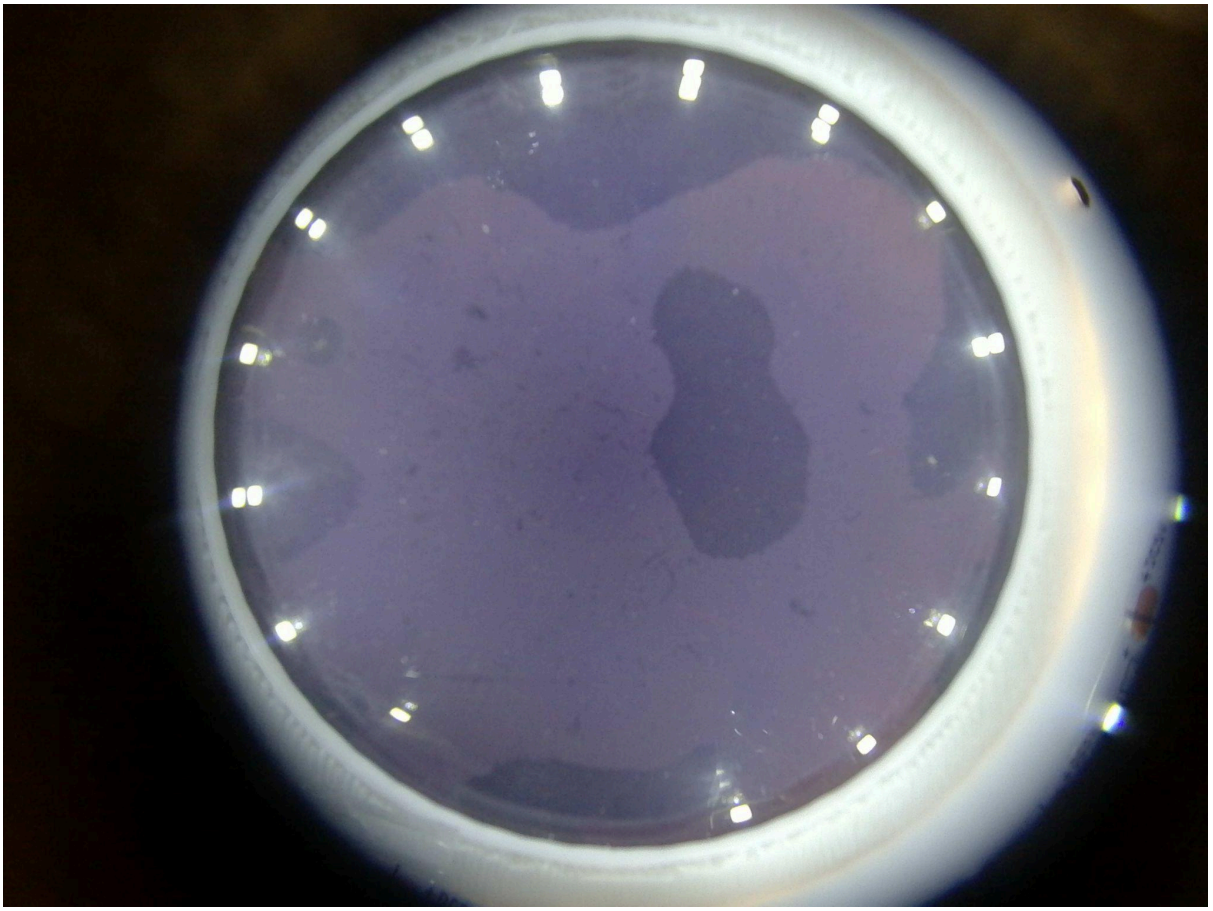
White Top, Black Core, LED strip a bit further down, anti-reflective shield, diffuse paper lining the interior at a few millimeters from light source

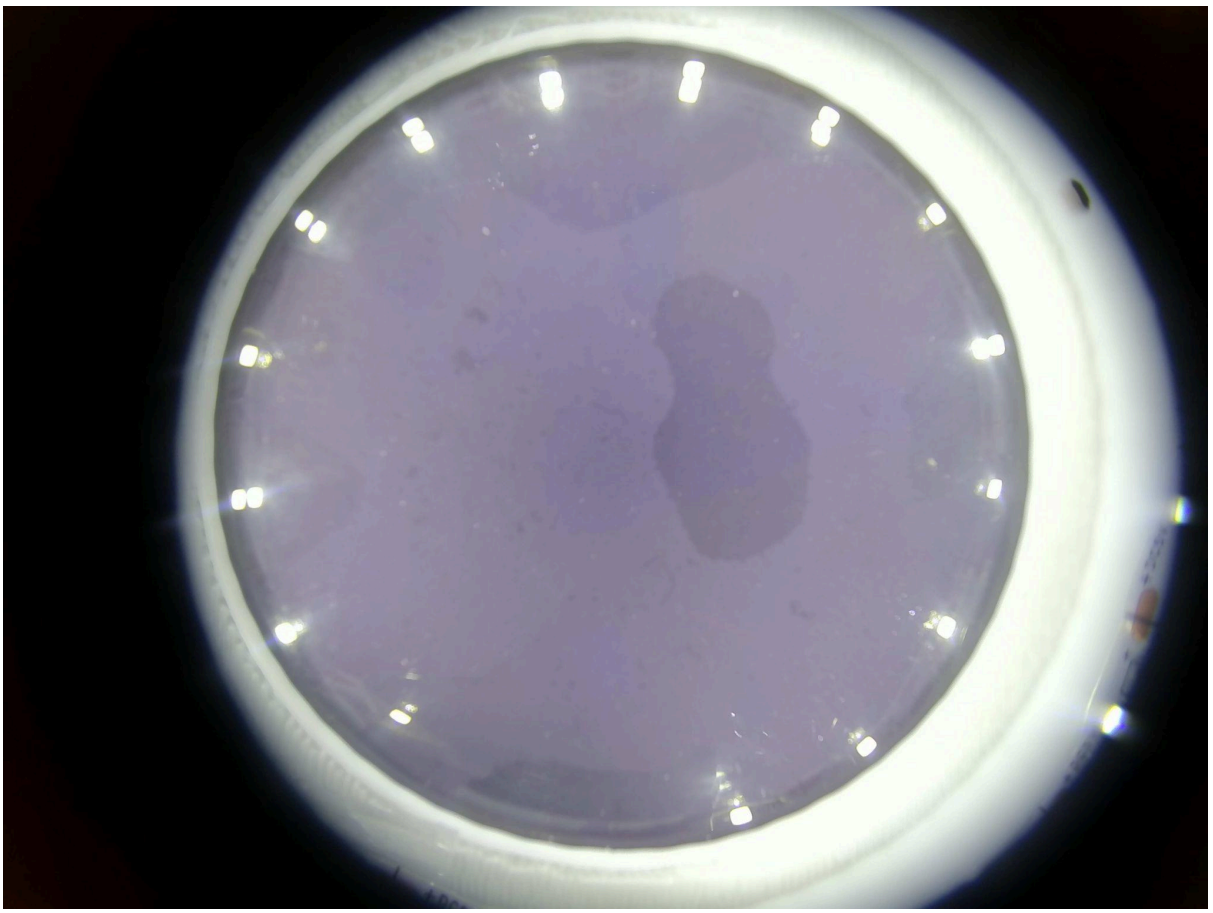


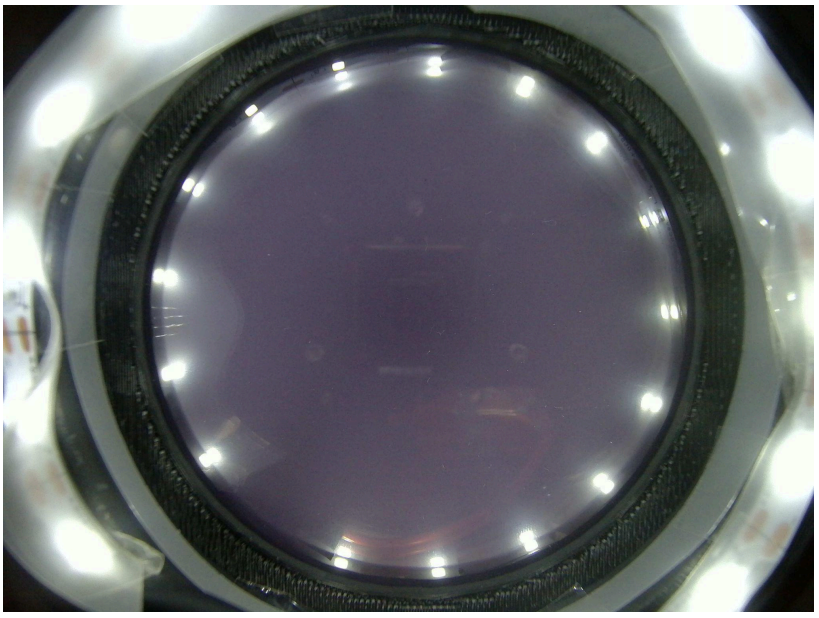
Black Top, Black Core, LED strip a bit further down, anti-reflective shield



Old attempts below



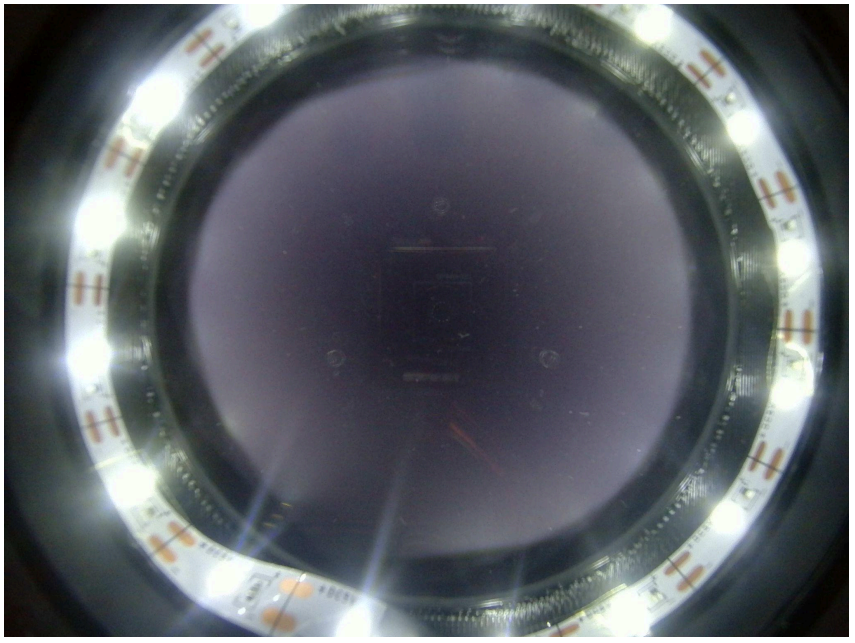




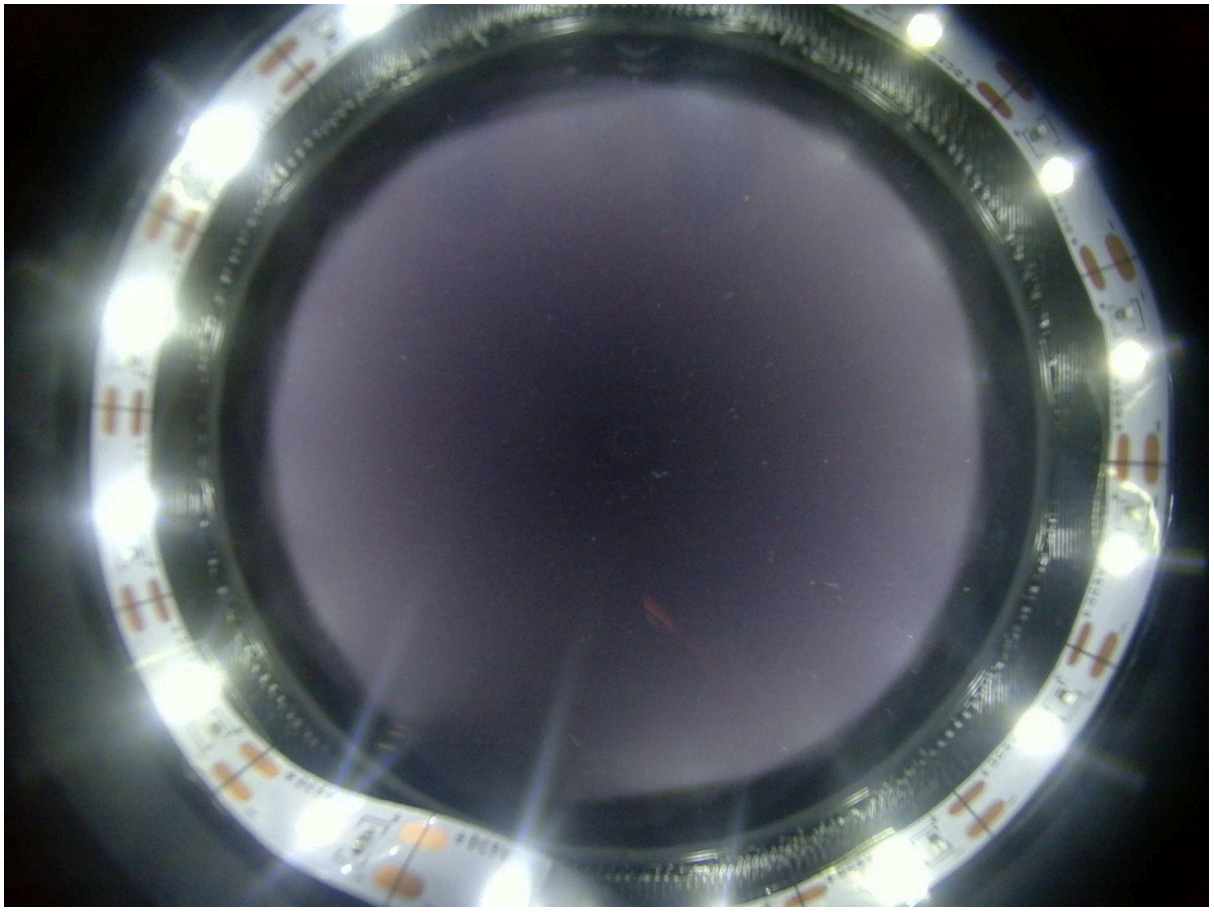
Initially, it looked a bit like this where the lights infringe on the surface and therefore make the region of interest smaller. There is also the issue of the reflection of the camera amongst other things that is visible.

There are two things that can be done to fix that, one is to put an anti reflective (black) shield. The second is to move the led strip higher.

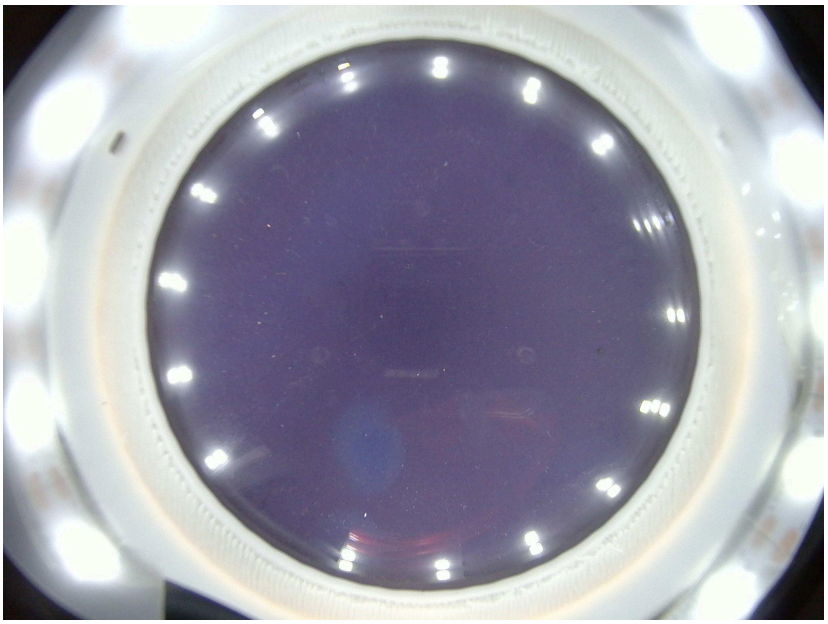
Moving the led strip higher gives the following:

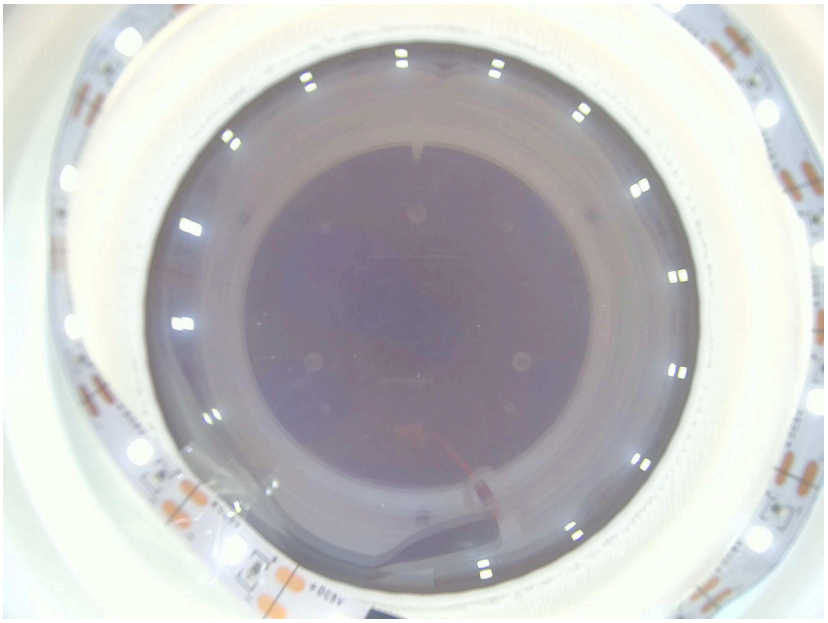


When an anti reflective shield is then added, we get the following:



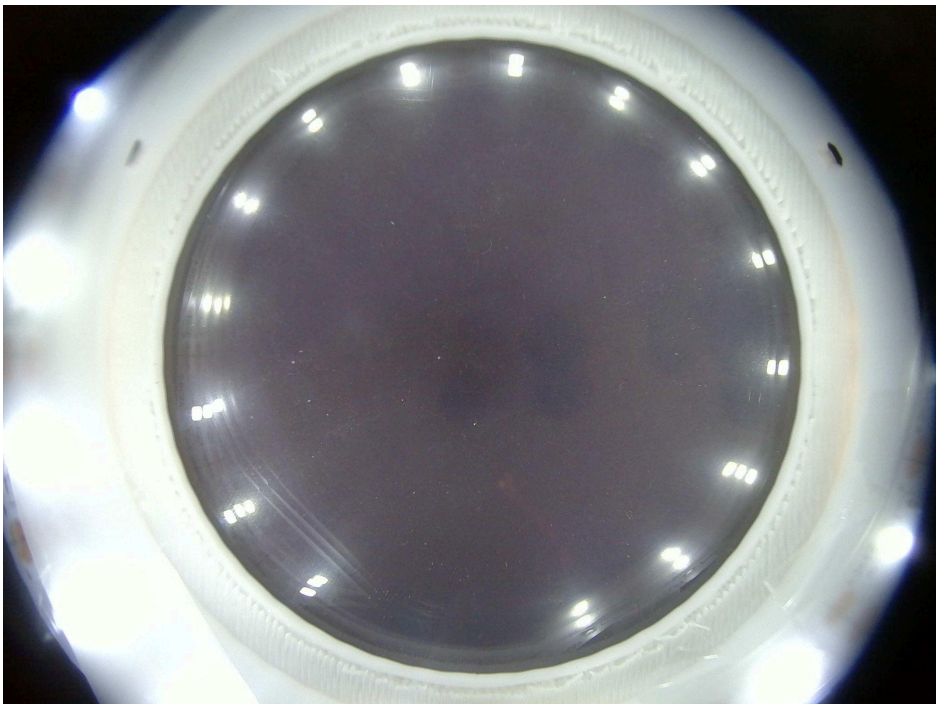
In order to increase the reflectiveness and try and make the illumination more consistent, I added white 3D printed pieces to the setup (with leds back in the original position and no anti reflective shield) and obtained the following:



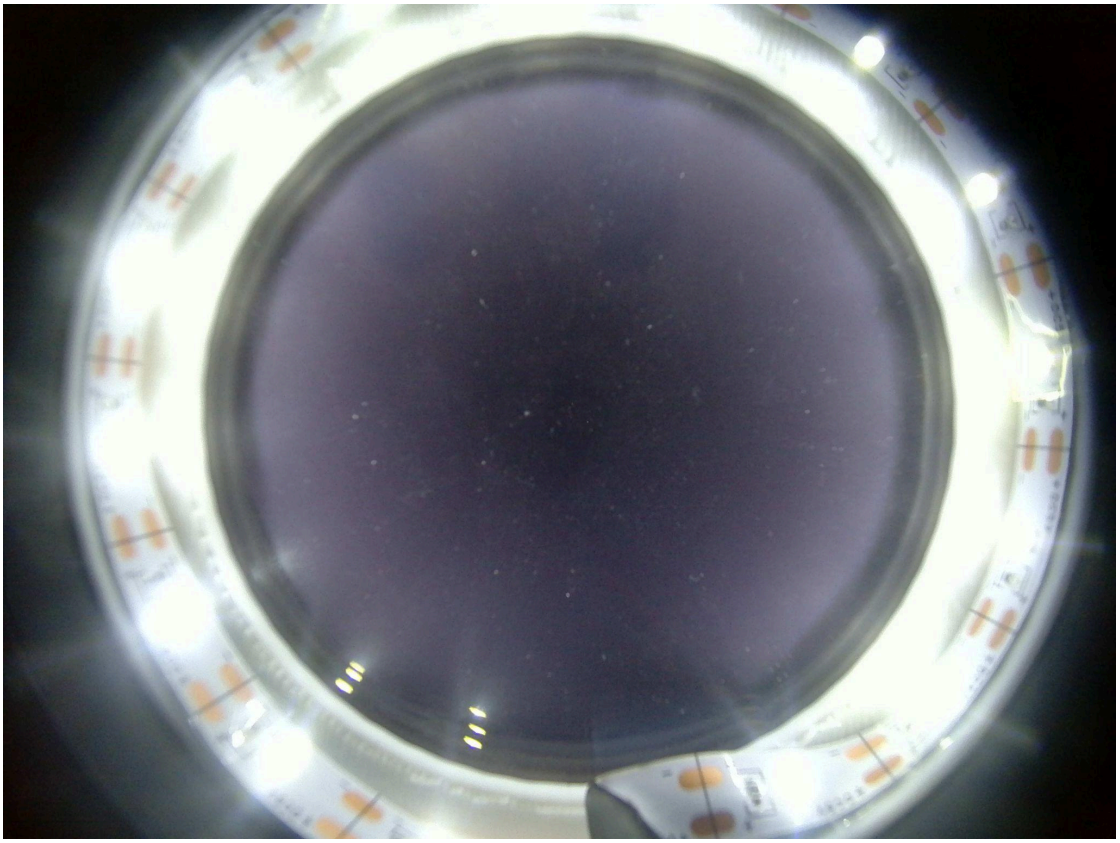


As one can see, the first image shows a clearer picture where only the top part is white. In the second picture, the entire setup is white and ends up being too reflective where it is hard to see the silicone.

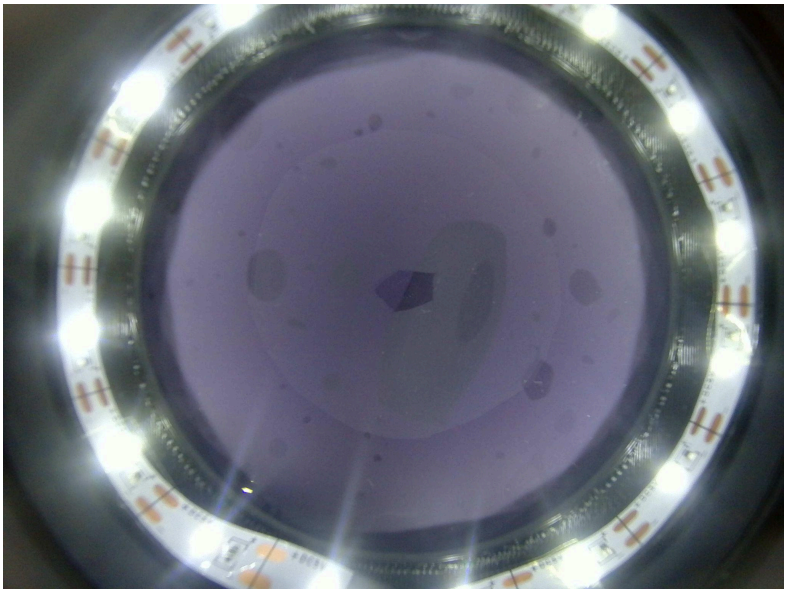
When just the top part is white, and with an anti reflective shield, the image becomes:

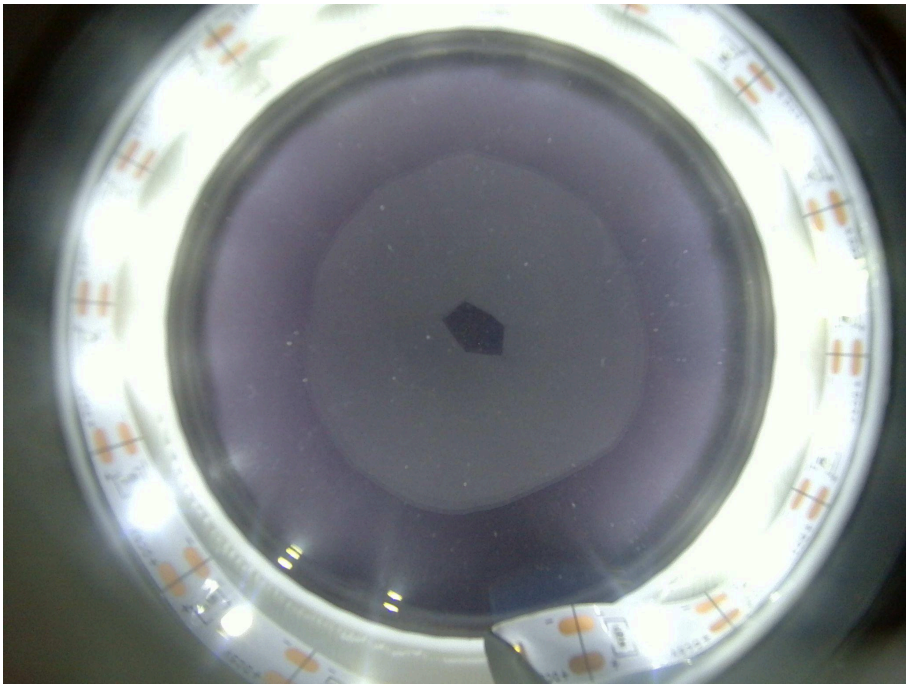


Moving the led strip higher with the same configuration (white top and anti reflective shield) gives:



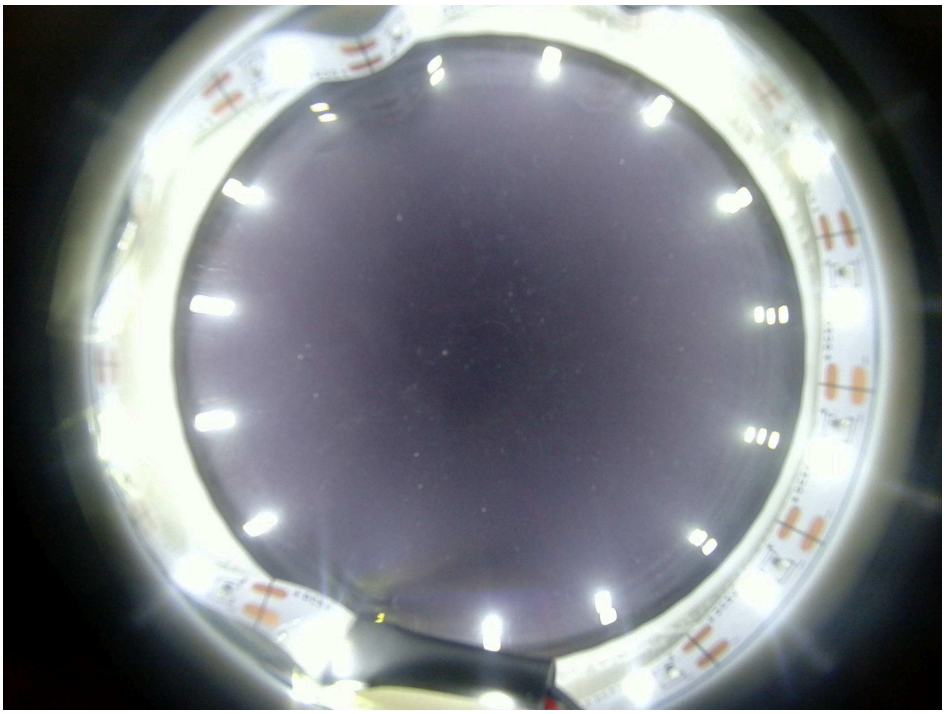
I then attempted to see in the cases of a white and black top what would happen if I replaced the anti-reflective shield with white paper, here are the results:





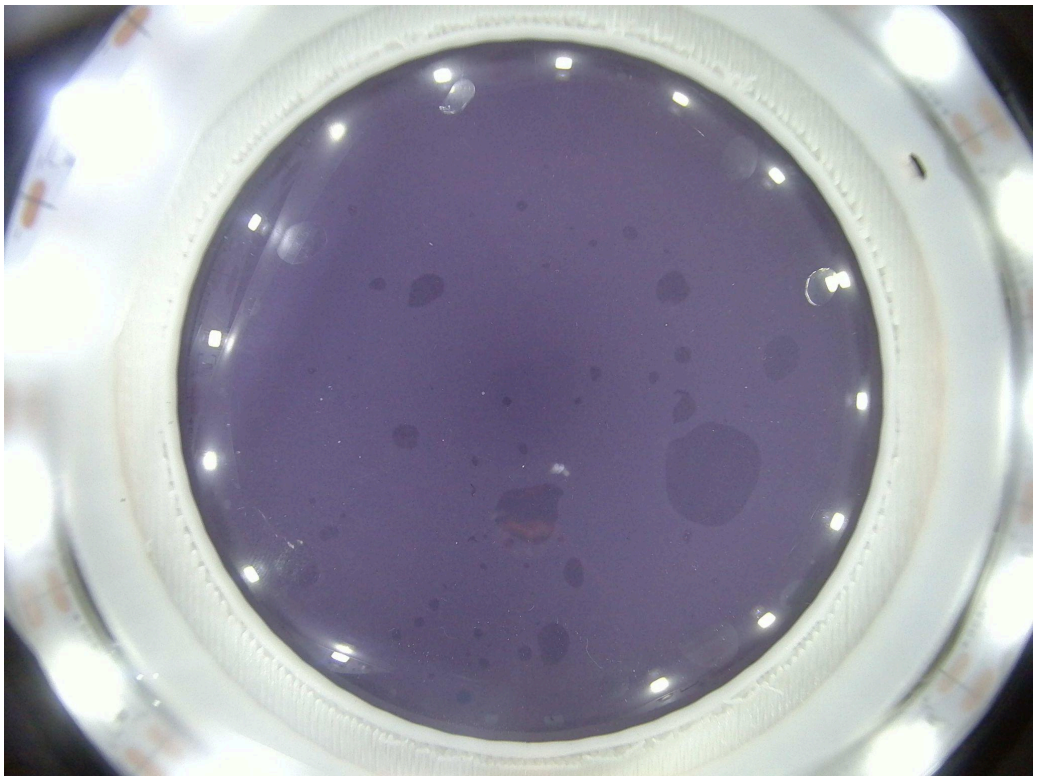
In both cases the paper is reflected and seen in the image which we do not wish, therefore I went back to the anti reflective shield.

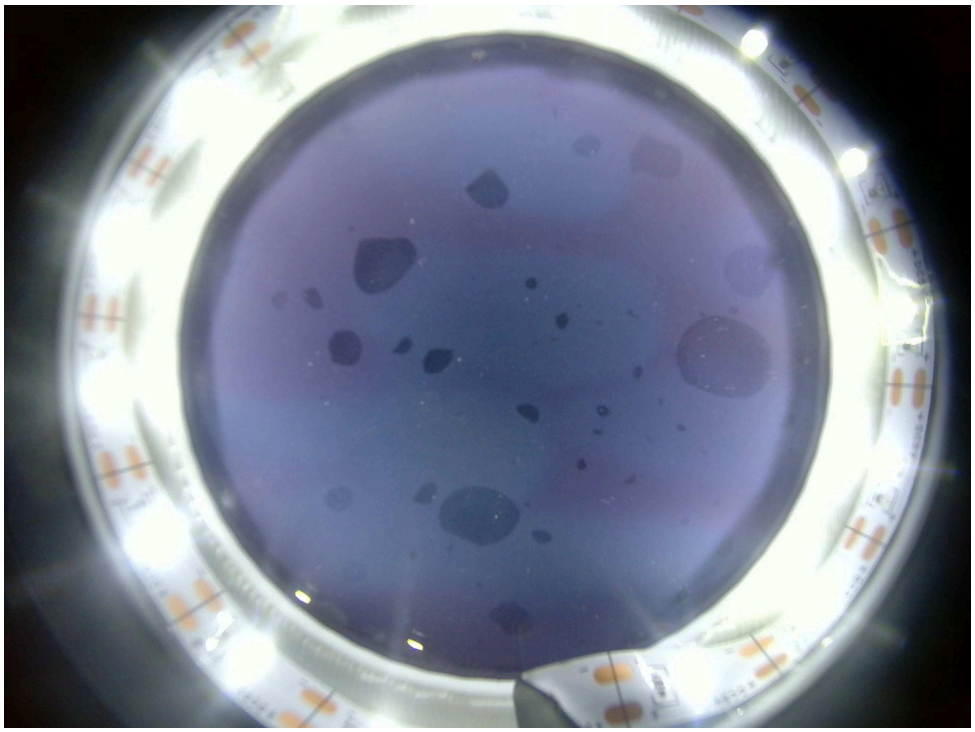
I then also tried to angle the lights (20°), seen below:



However, now the lights reappear within the frame which narrows the region of interest once more.

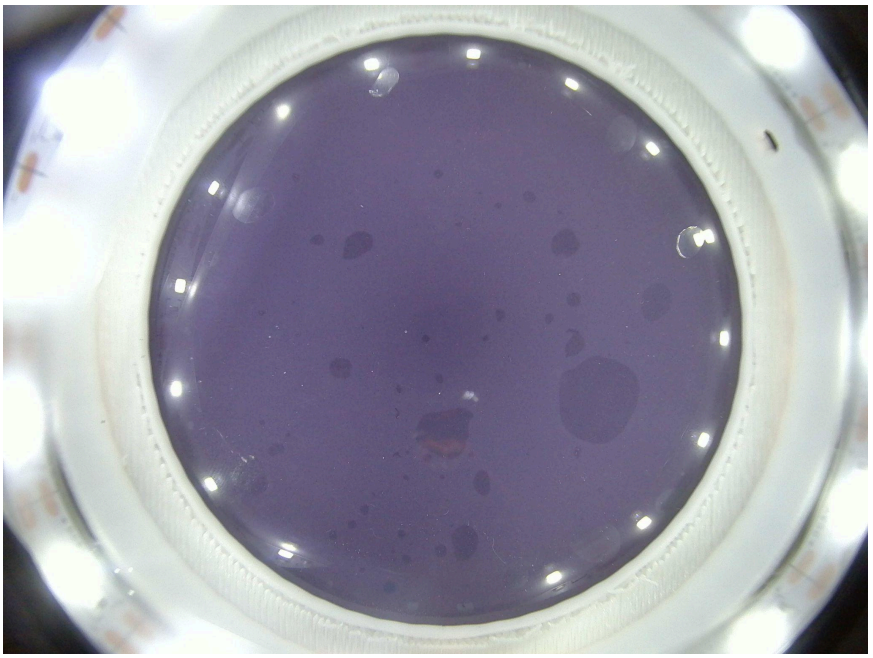
I also tried in a few instances what it looks like when the silicone appears directly on the glass (some air bubbles appear but a way could be found to remove them), here are some examples:

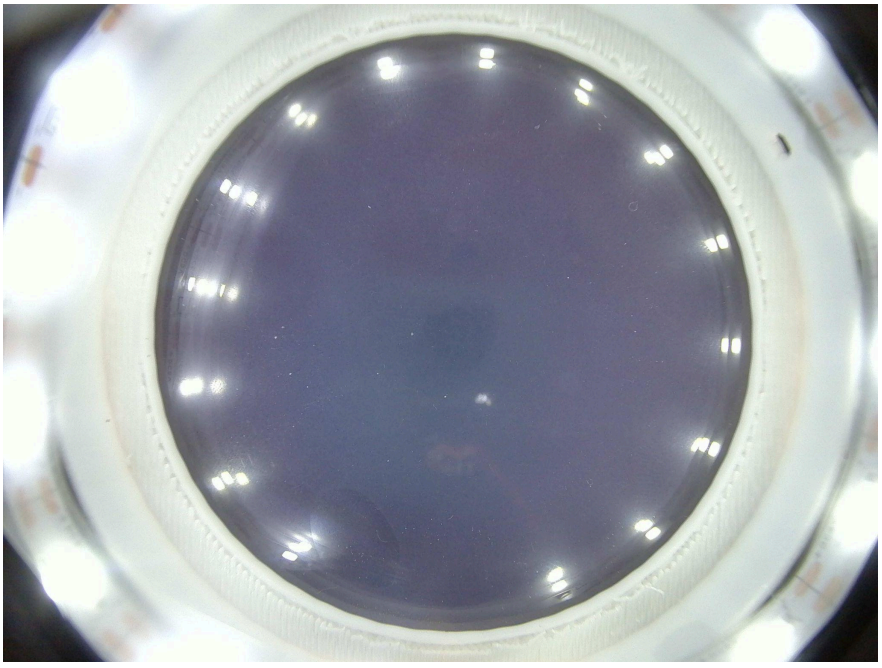


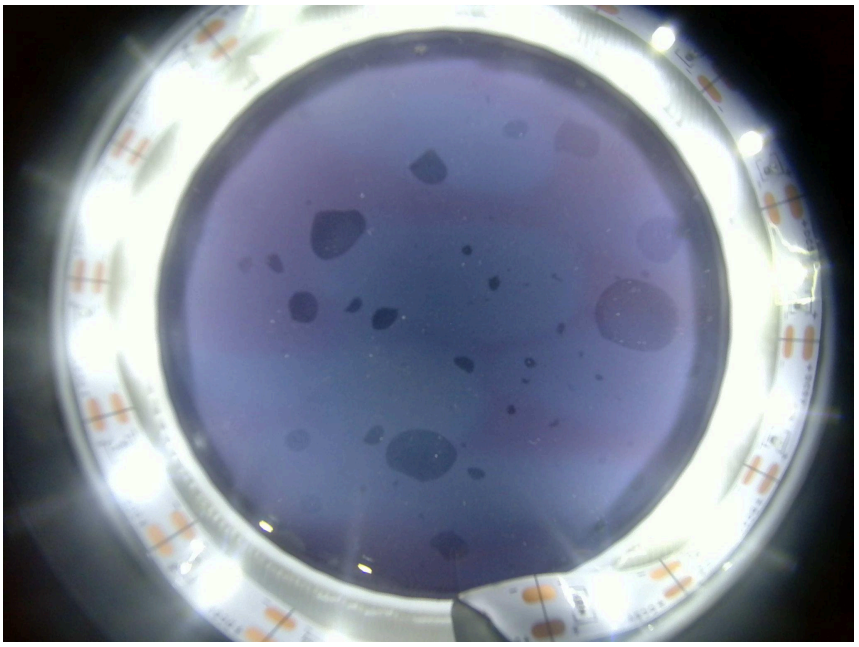


These appear to be a little bit better than when there is still the transparent sheet in between the silicone and the glass, however the bubbles would need to be removed somehow.

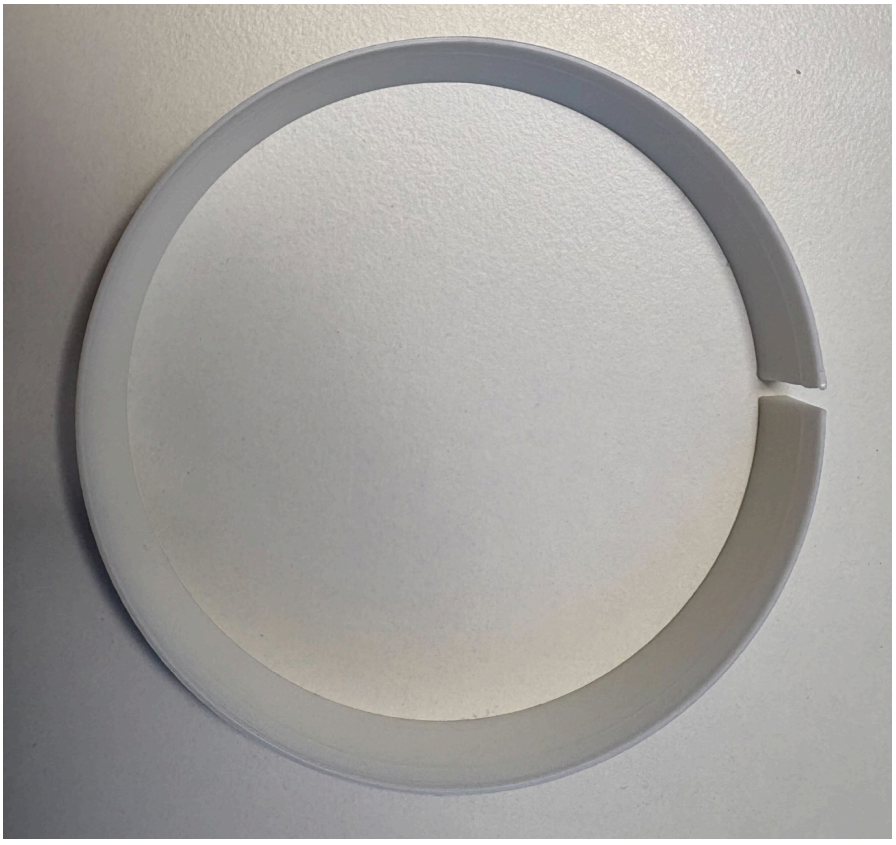
In my opinion, the better ones are the following:

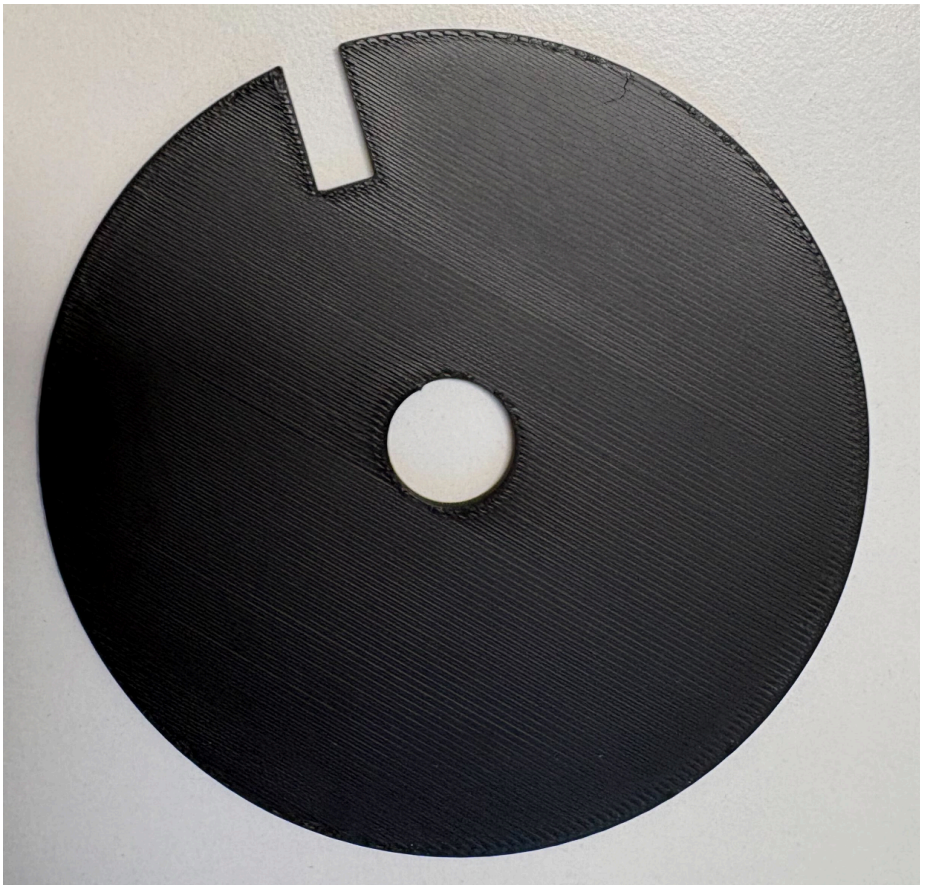






Below are two 3D printed parts that I used, the first part was used to angle the lights, and the second was the anti-reflective shield:





Self Notes

Psycho Paint → only for when using the screen printing with frame as a lot of solvent is added to it in order for it to pass through

40 on the adjustable film applicator → 0.1mm thickness
60 on the adjustable film applicator → 0.2mm thickness
100 on the adjustable film applicator → 0.5mm thickness

Plan for Grating Experiment

Black and White Grating Experiment Plan

Five different measurements to make:

1. Force
2. Depth
3. Shape
4. Stiffness
5. 3D reconstruction

Force-displacement curve

Force

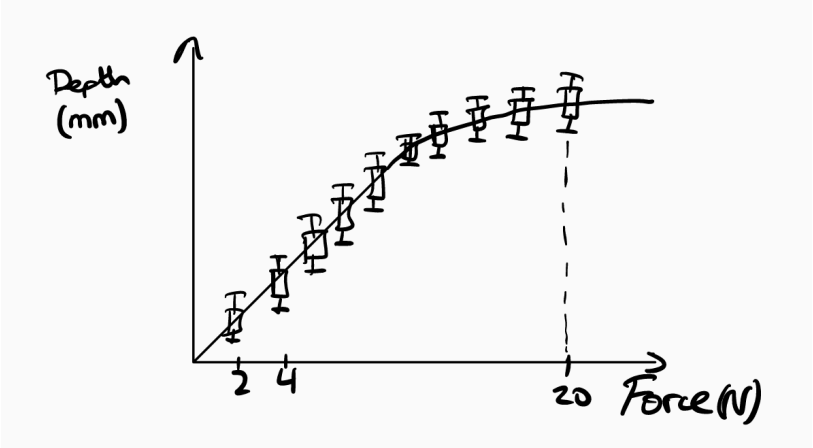
One indenter tip:

- Flat Circular

Do one pre-test to see maximal range (10N? 20N?)

Range: 0-48N
Steps: 2N → (2N, 4N, 6N, 8N, 10N, 12N, 14N, 16N, 18N, 20N)
Number of times to repeat: 5
Number of datapoints: 10x5 = 50

Desired Graph:



Depth

One indenter tip:

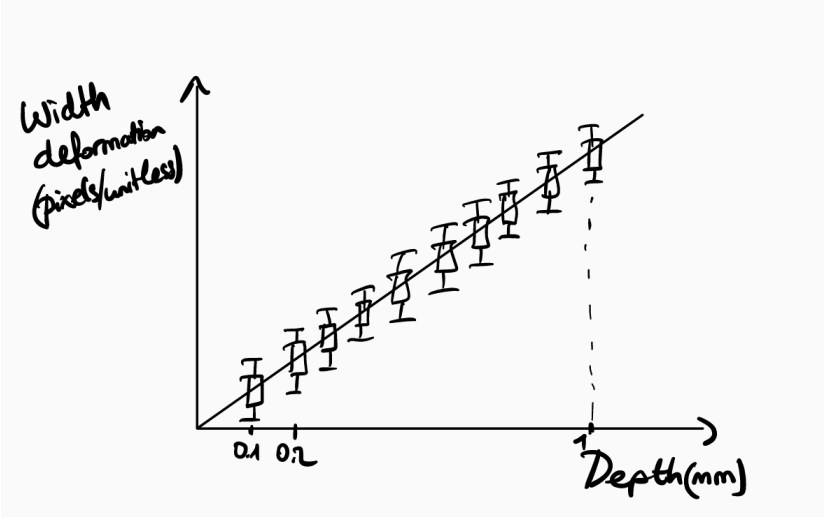
- Flat Circular

To see max depth → Find out from Force graph

Change: make it force vs width

Range: 0-3.20mm
Steps: 0.2mm → (0.2mm, 0.4mm, 0.6mm, 0.8mm, 1.0mm, 1.2mm, 1.4mm, 1.6mm, 1.8mm, 2.0mm)
Number of times to repeat: 5
Number of datapoints: 16x5 = 80

Desired Graph:



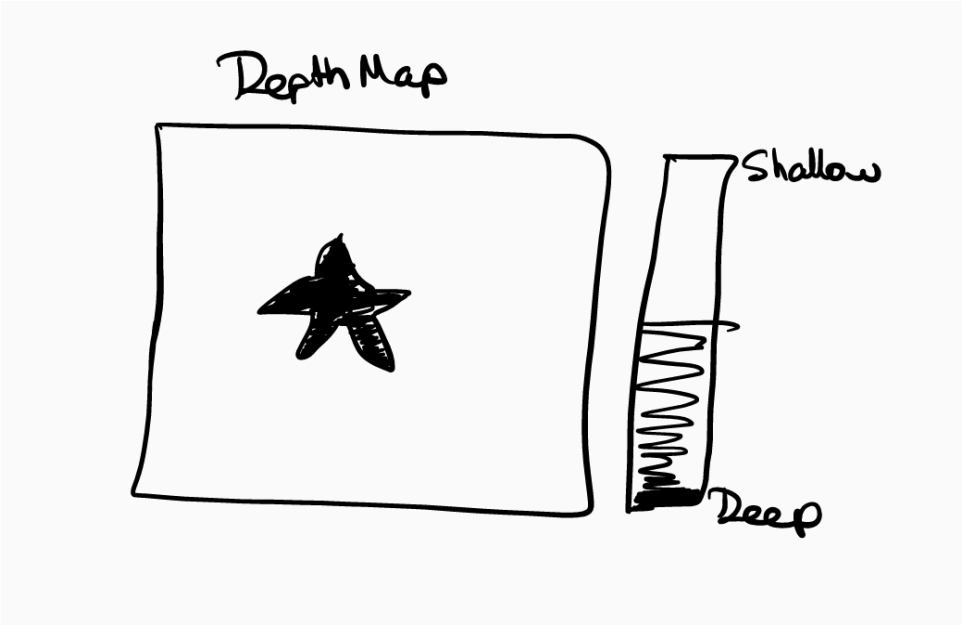
Shape

Three indenter tips:

- Star
- Ring (must reprint slightly larger)
- 3 circles

Constant Depth: 1mm
Number of times to repeat: 5
Number of datapoints: 3x5 = 15

Desired Graph:



Stiffness

One indenter tips:

- Flat Circular

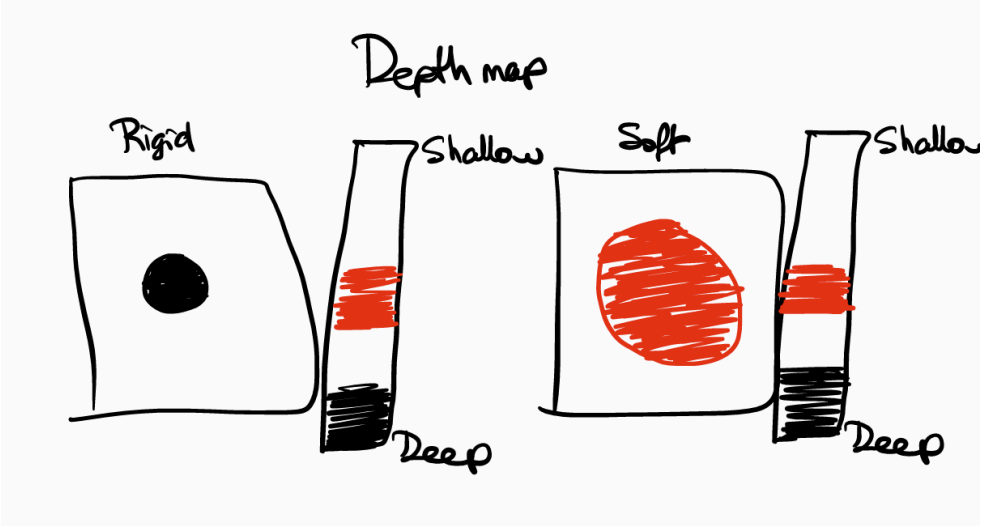
Two materials:

- PLA (Rigid)
- TPU (Soft) (to be made and 3D printed)
- DragonSkin10/20?

Constant Force: 10N

Number of times to repeat: 5
Number of datapoints: 2x5 = 10

Desired Graph (soft tip does not go as deep but expands over larger surface area):



3D Reconstruction

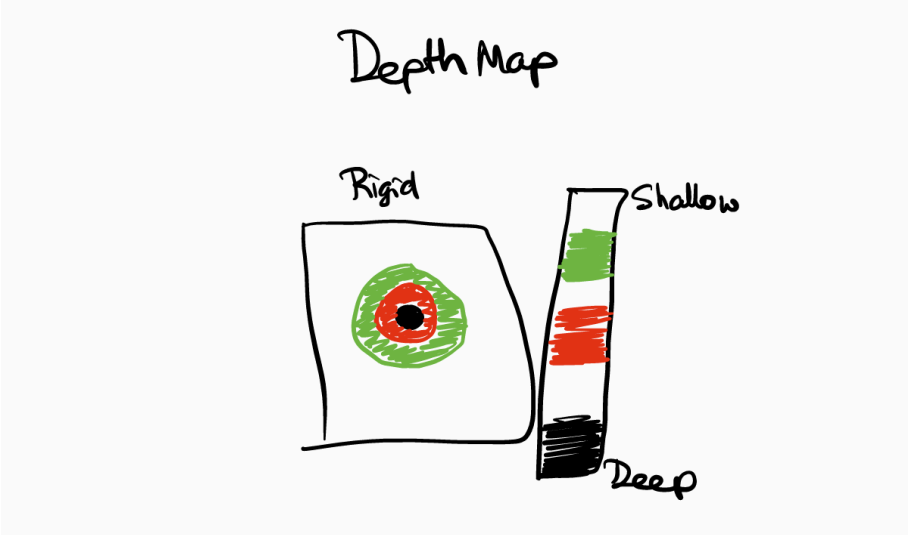
Two indenter tips:

- Hemisphere
- Starfish
- Mountain top (to be made and 3D printed)

Constant Depth: 1mm

Number of times to repeat: 5
Number of datapoints: 3x5 = 15

Desired Graph:



Old
Known indentation → important for 3D reconstruction

Want 5 different indenters:

- Flat circular
- Star
- Spherical
- 3 circles
- Ring

Having these 4 different indenters allows to have different for different aspects, like the star which is a different shape, the sphere that has different depths, and the 3 circles that looks for separate deformations in the same image, and the ring where there is a hole in the middle

0.0 mm set to when indenter just makes contact with the grating
Then do 10 measurements → 0.10mm, 0.20mm, 0.30mm, 0.40mm, 0.50mm, 0.60mm, 0.70mm, 0.80mm, 0.90mm, 1.00mm
Maybe more if elastic range allows for it

Take 5 pictures for data

Make indentation folder, and then within each folder have a sub folder with all the measurement values (so folder name: 010, 020) that contains the pictures

From there → develop algorithm to look at grating shift (outline of shape) and match with width of grating (depth of indentation) to try and get 3D reconstruction

Steps:

1. Take first indenter and attach to machine
2. Calibrate the 0.0mm to when indenter just touches the very top of the grating
3. Then set the machine to desired measurement (0.10mm, 0.20mm, 0.30mm, 0.40mm, 0.50mm, 0.60mm, 0.70mm, 0.80mm, 0.90mm, 1.00mm)
4. Once it reaches the desired measurement, wait a few seconds and then take 5 pictures
5. Repeat steps 3 & 4 until pictures of all desired measurements have been taken
6. Then switch out the indenter to the new indenter and then repeat steps 2-5 until all desired indenters have been used

Known Force

Use the same 5 indenters:

- Flat circular
- Star
- Spherical
- 3 circles
- Ring

Determine what the max force to apply is (start with 10N) and then decide on even steps within that:
1N, 2N, 3N, 4N, 5N, 6N, 7N, 8N, 9N, 10N to start off with

Should even be able to go up to 20N → can do a second run of 11N, 12N, 13N, 14N, 15N, 16N, 17N, 18N, 19N, 20N

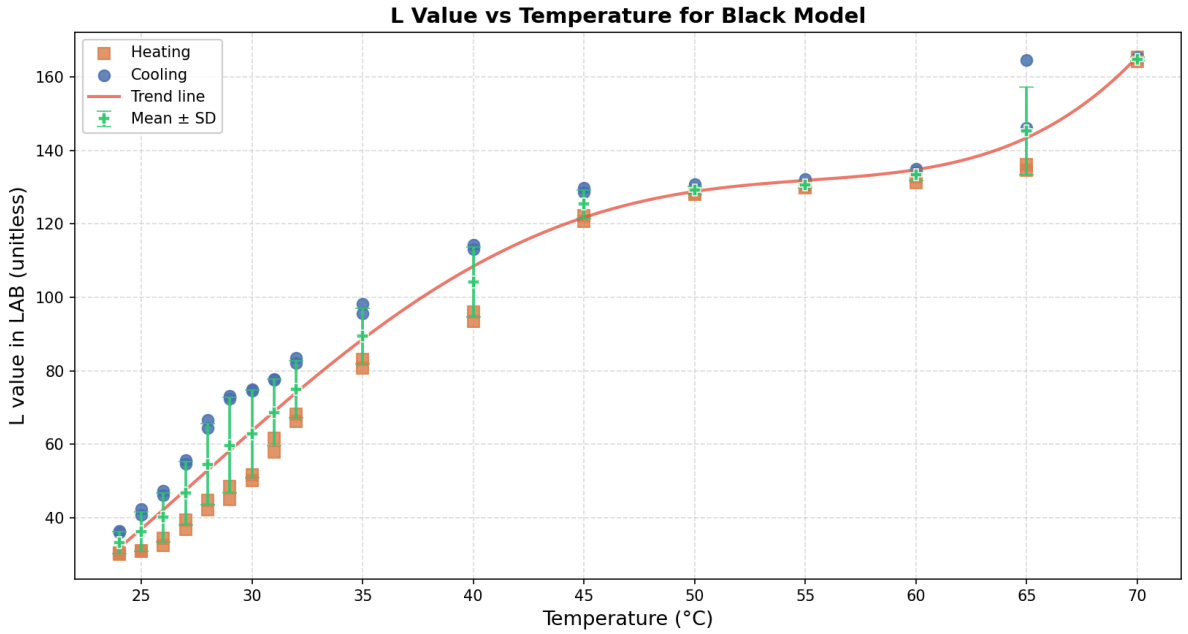
Take 5 pictures for data

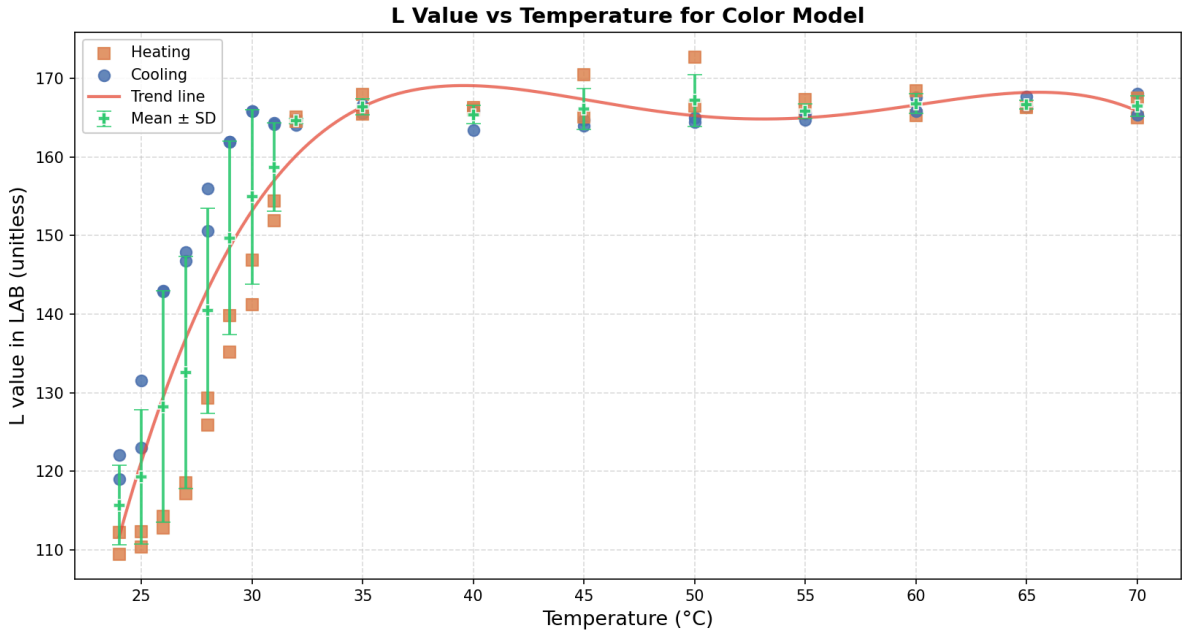
Make force folder, and then within each folder have a sub folder with all the measurement values (so folder name: 1, 2, etc…) that contains the pictures

Steps:

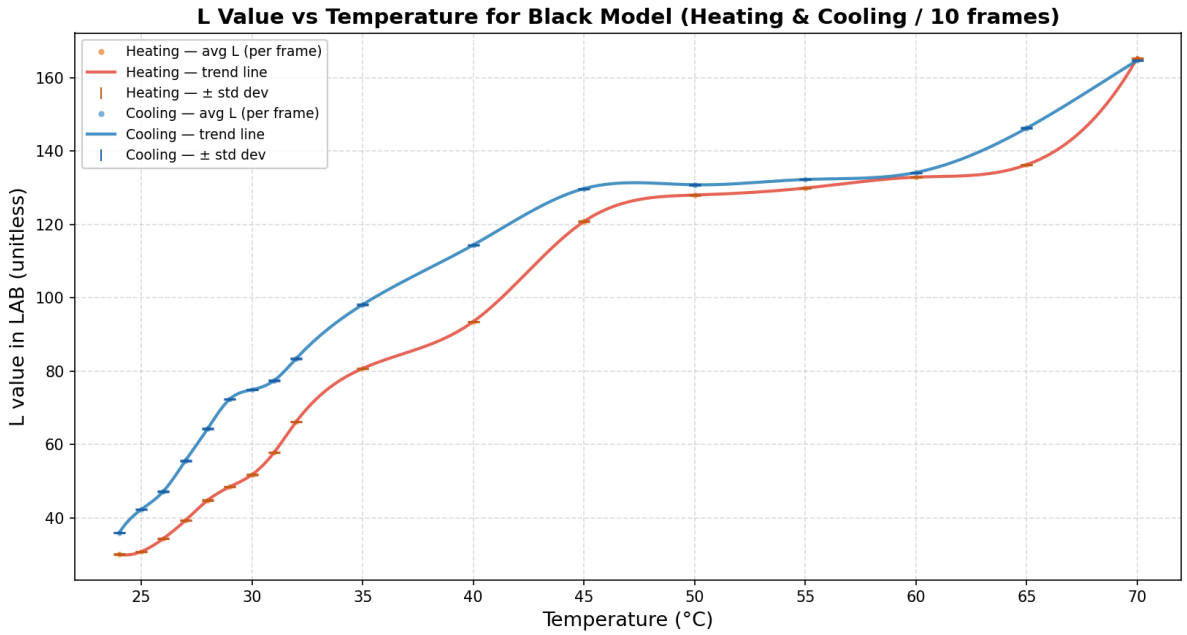
1. Take first indenter and attach to machine
2. Calibrate the 0.0mm to when indenter just touches the very top of the grating
3. Then set the machine to desired force (1N, 2N, 3N, 4N, 5N, 6N, 7N, 8N, 9N, 10N)
4. Once it reaches the desired force, wait a few seconds and then take 5 pictures
5. Repeat steps 3 & 4 until pictures of all desired forces have been taken
6. Then switch out the indenter to the new indenter and then repeat steps 2-5 until all desired indenters have been used

Plots

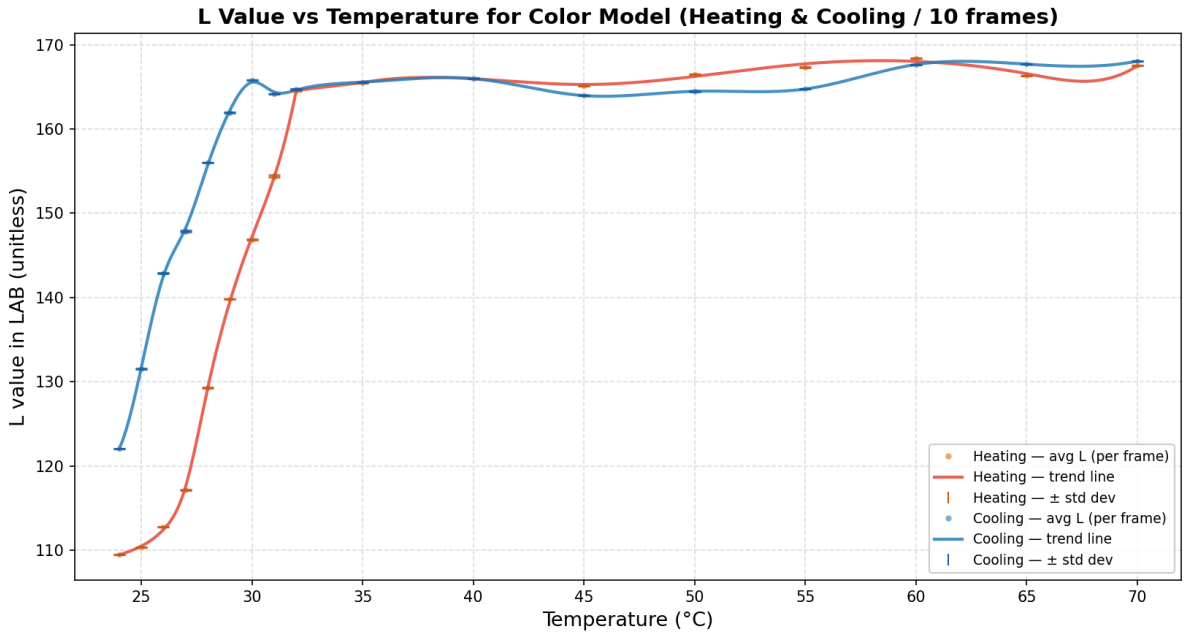

L Value vs Temperature for Black Model
Heating
Cooling
Trend line
Mean ± SD
L value in LAB (unitless)
Temperature (°C)

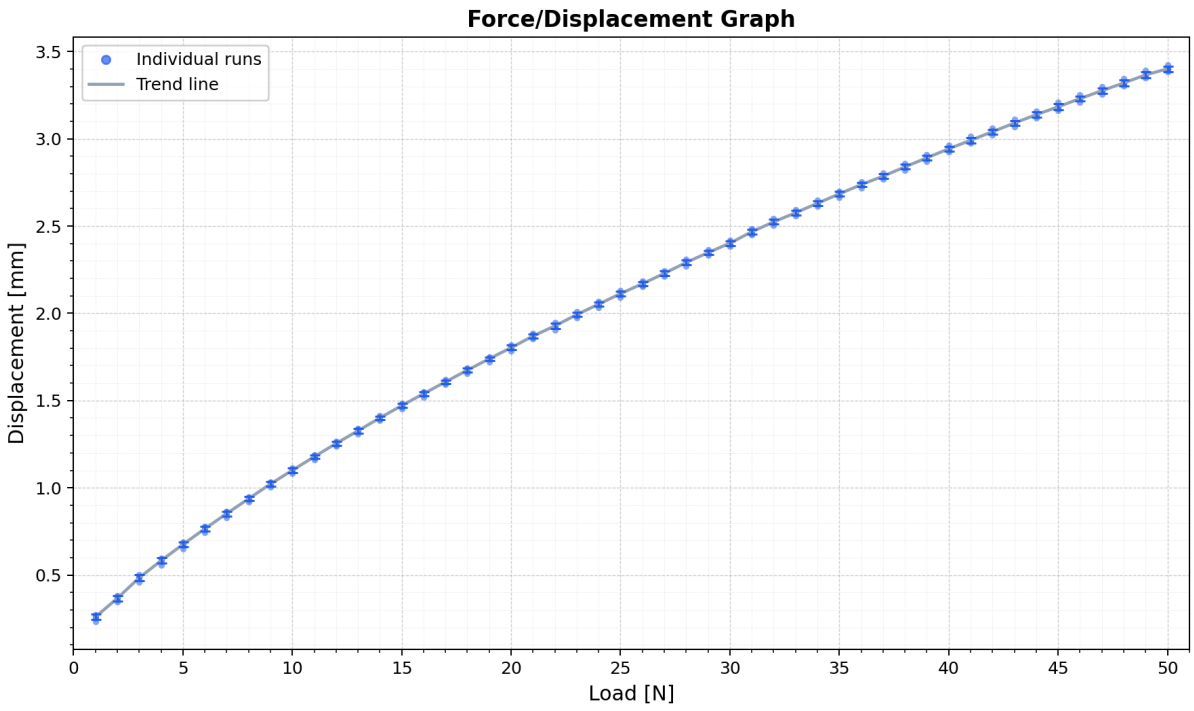

L Value vs Temperature for Color Model
Heating
Cooling
Trend line
Mean ± SD
L value in LAB (unitless)
Temperature (°C)

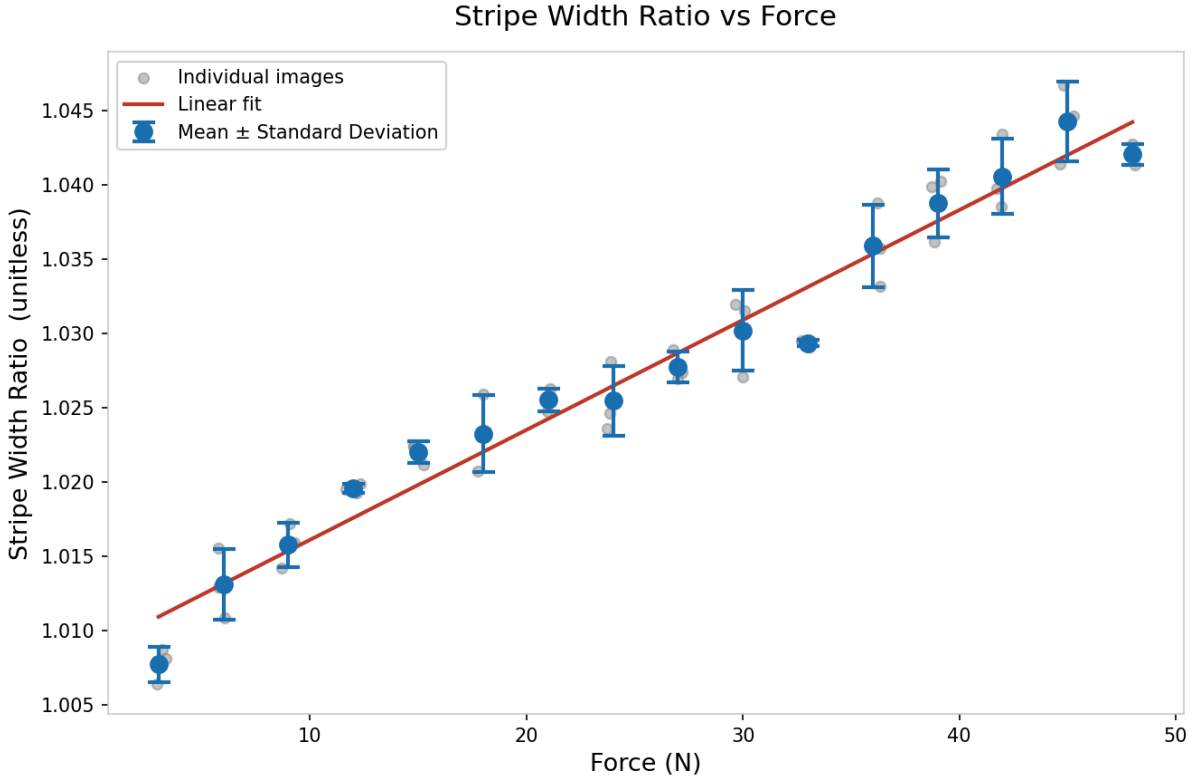
Black heating and cooling - run 1



Color mixed heating - 31 deg C from run 1, rest from run 2
Color mixed cooling - 35 deg C from run 2, rest from run 1



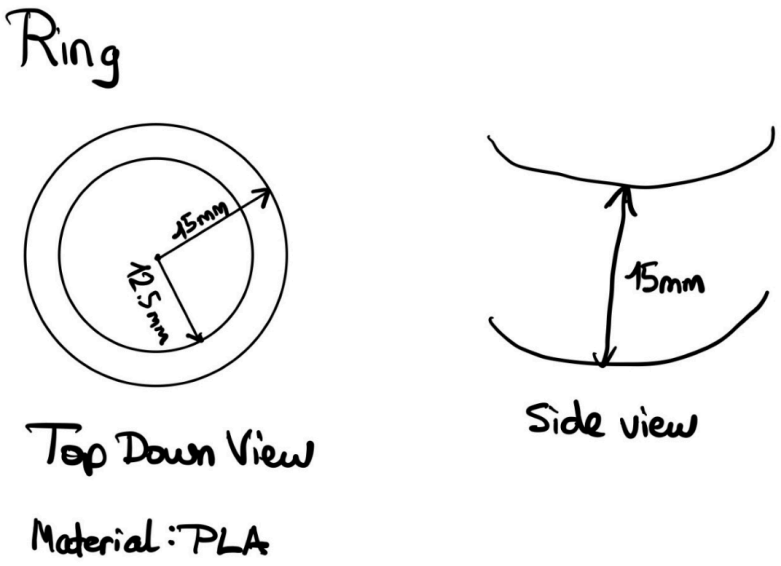

Force/Displacement Graph
Individual runs
Trend line
Displacement [mm]
Load [N]
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
5
10
15
20
25
30
35
40
45
50


Stripe Width Ratio vs Force
Individual images
Linear fit
Mean ± Standard Deviation
Stripe Width Ratio (unitless)
Force (N)
1.005
1.010
1.015
1.020
1.025
1.030
1.035
1.040
1.045
10
20
30
40
50

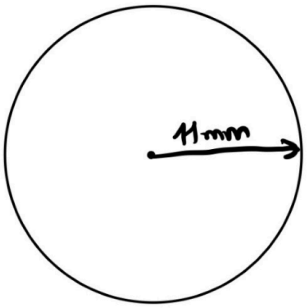
Tips to be 3D printed

Tips to be printed:

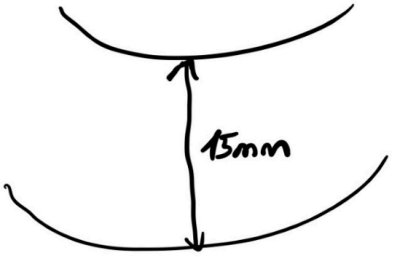
- Ring (PLA)
- Flat Circle (PLA, TPU, mold for DragonSkin)
- Flat Star (PLA)
- multiple poionts (PLA)
- Hemisphere (PLA)
- Starfish (PLA)



Flat Circle



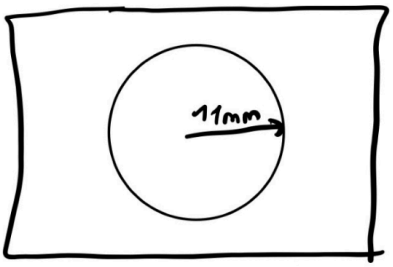
Top Down View



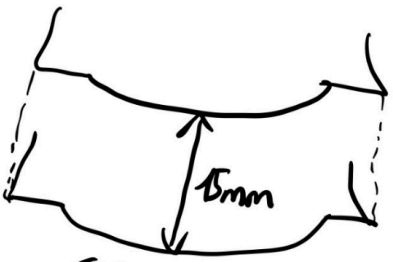
Side View

Material: TPU

Flat Circle Mold (for Dragonskin)



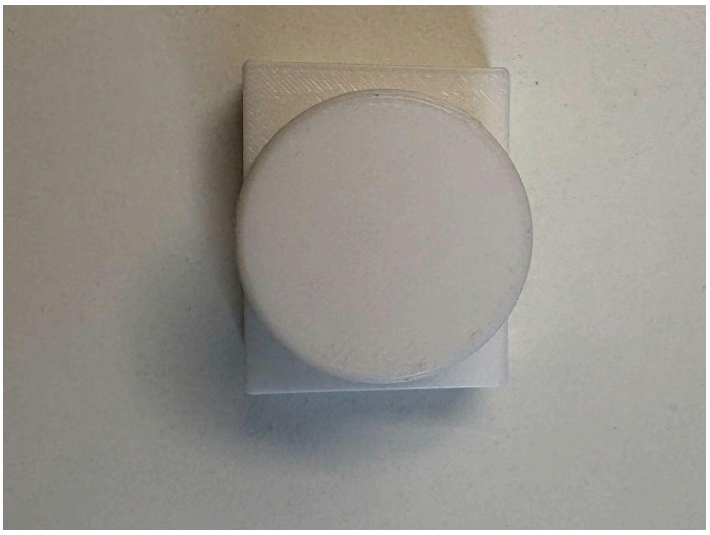
Top Down View



Side View

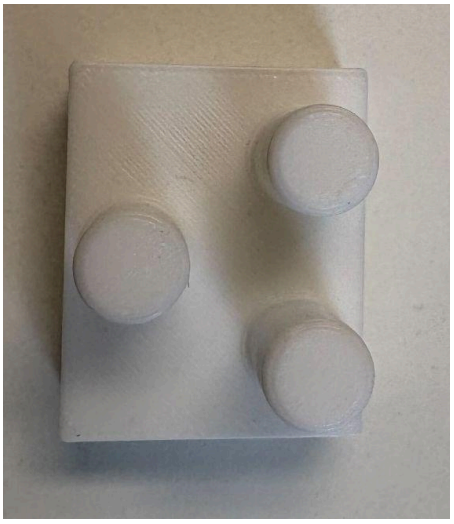
Material: PLA

Flat Circle, material: PLA

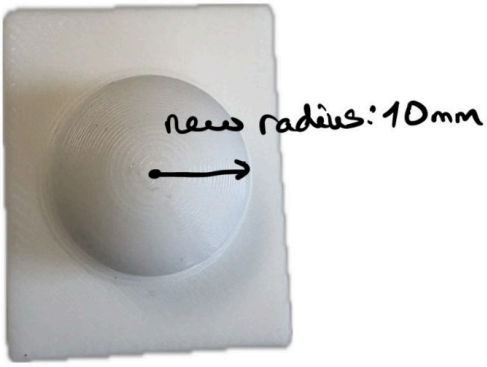


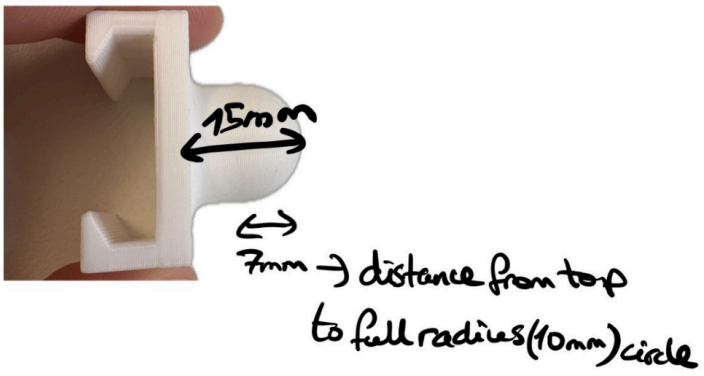
Flat Star, material: PLA

Three Circles, material: PLA

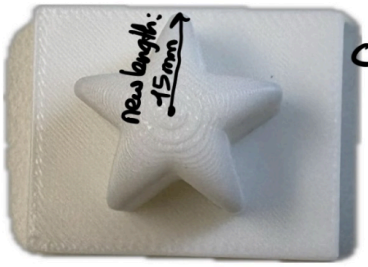


Hemisphere, material: PLA

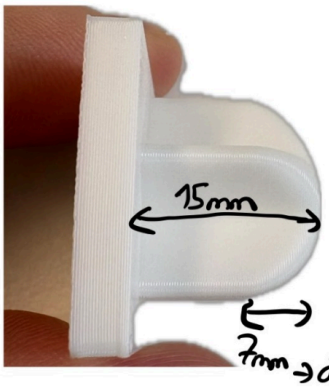




Starfish, material: PLA



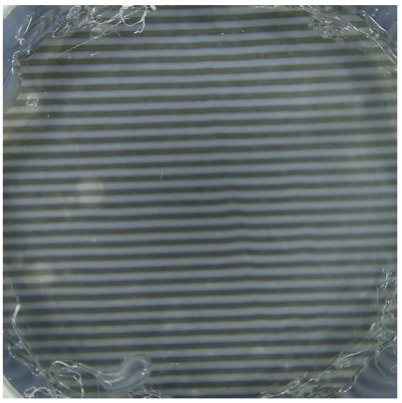
center to all edges: 15mm



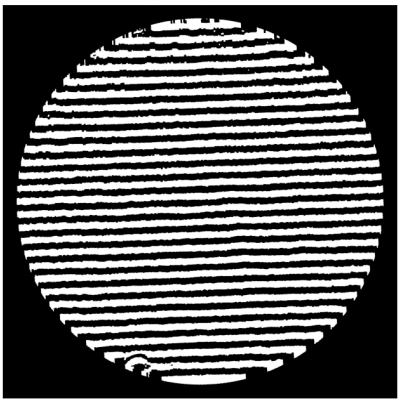
→ distance from top to full star

Stripe Segmentation

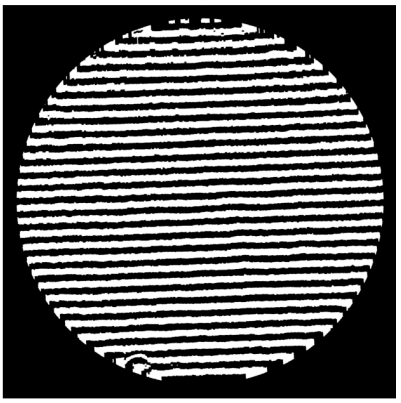
Original



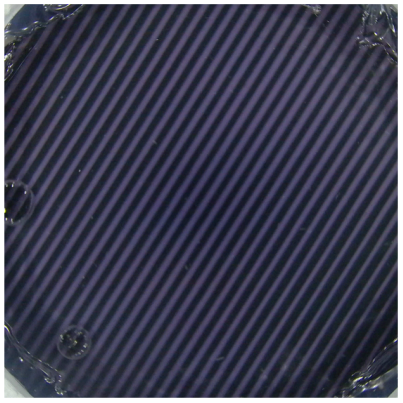
Bright stripe mask



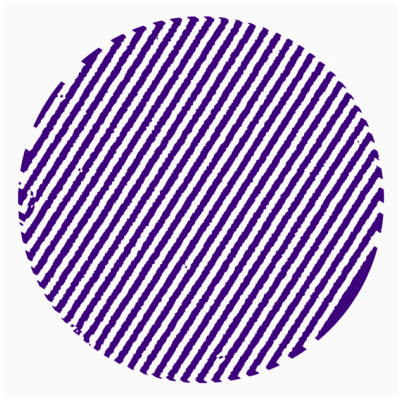
Dark stripe mask



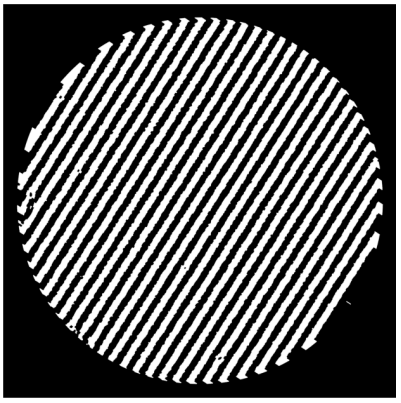
Original



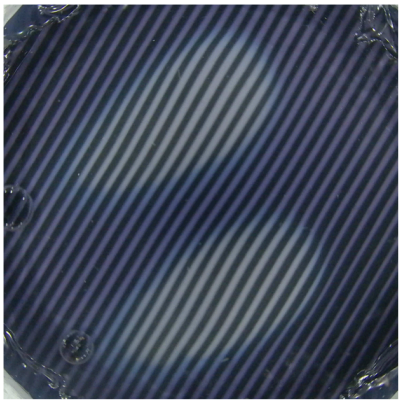
Purple mask



Black mask

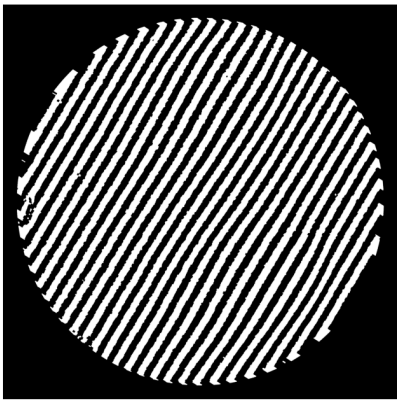


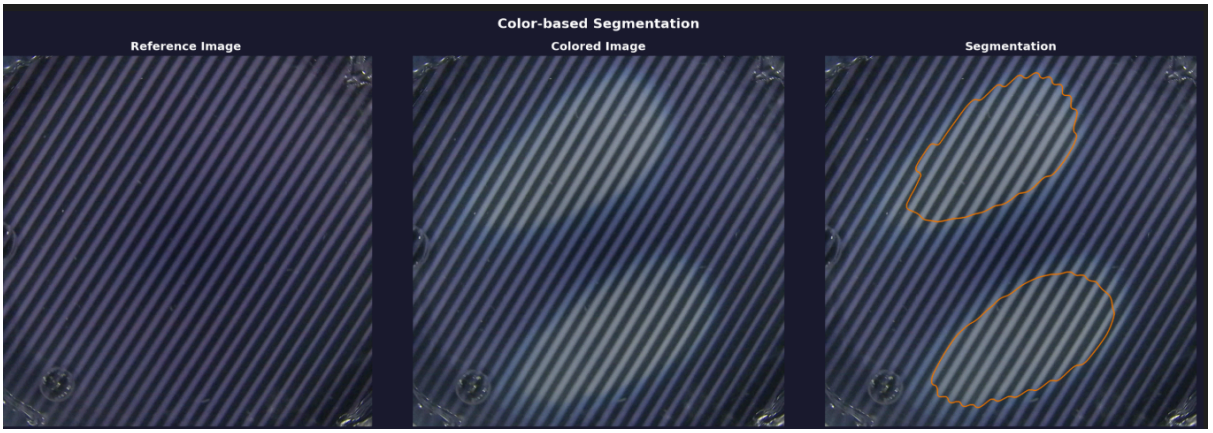
Original



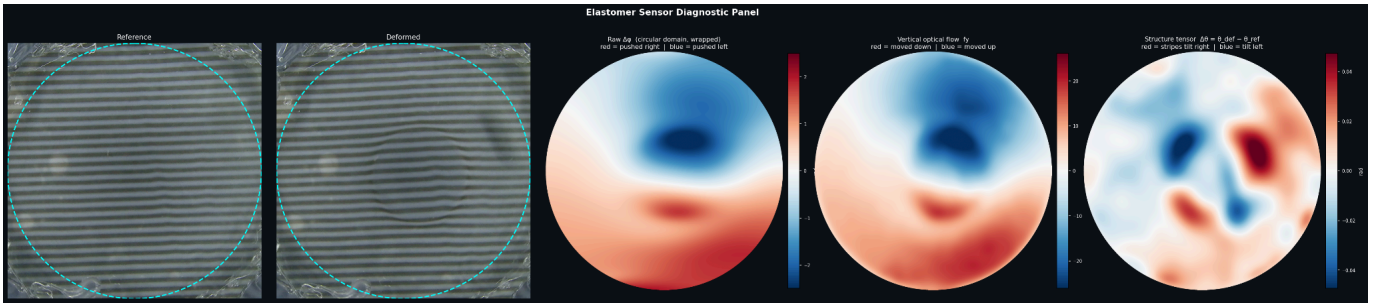
Purple mask

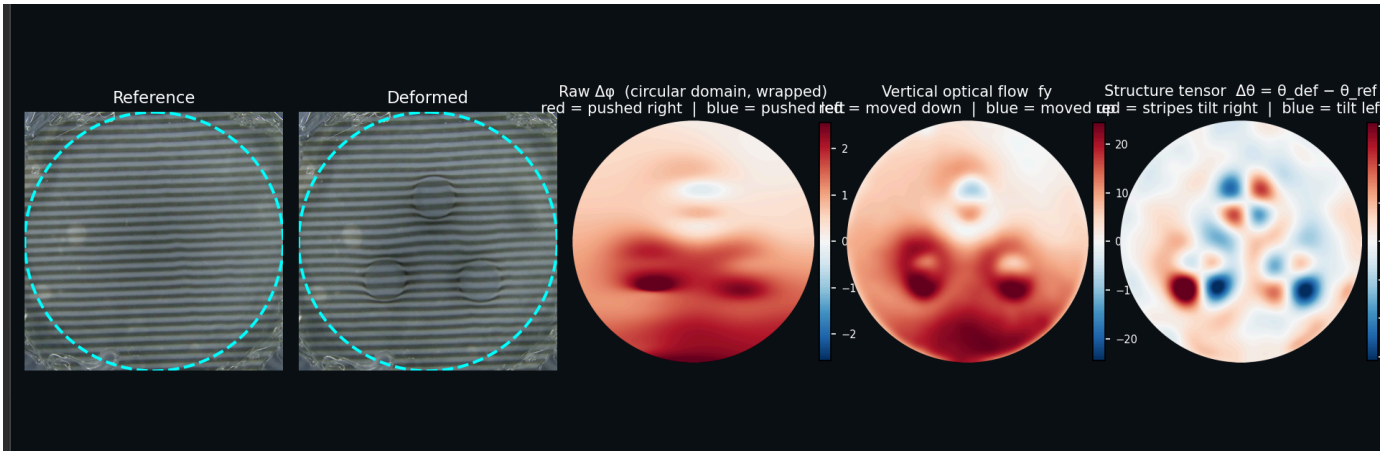
Black mask

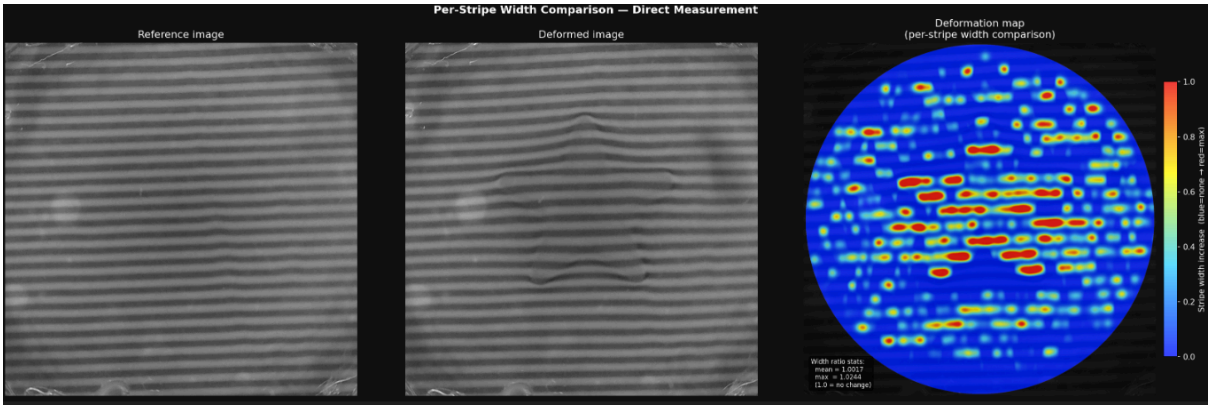


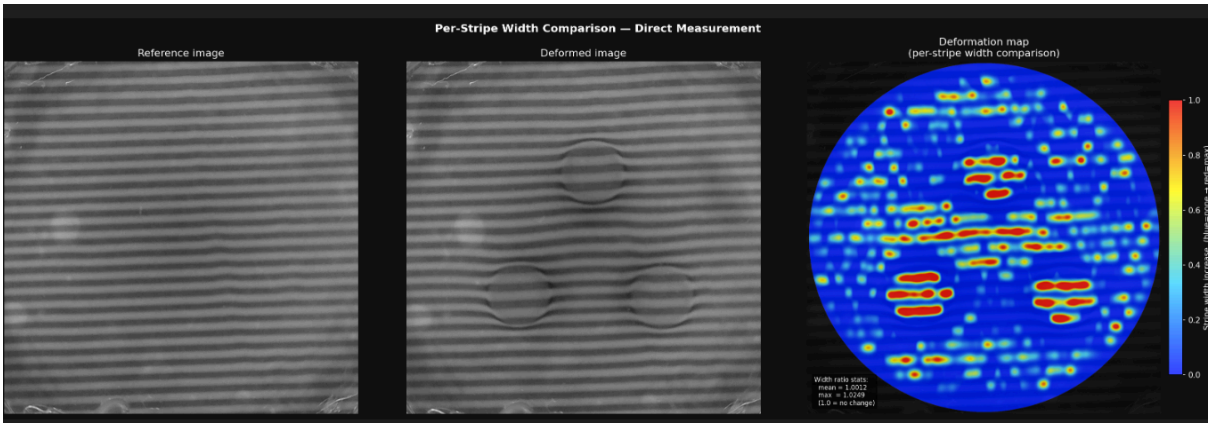

Color-based Segmentation
Reference Image
Colored Image
Segmentation

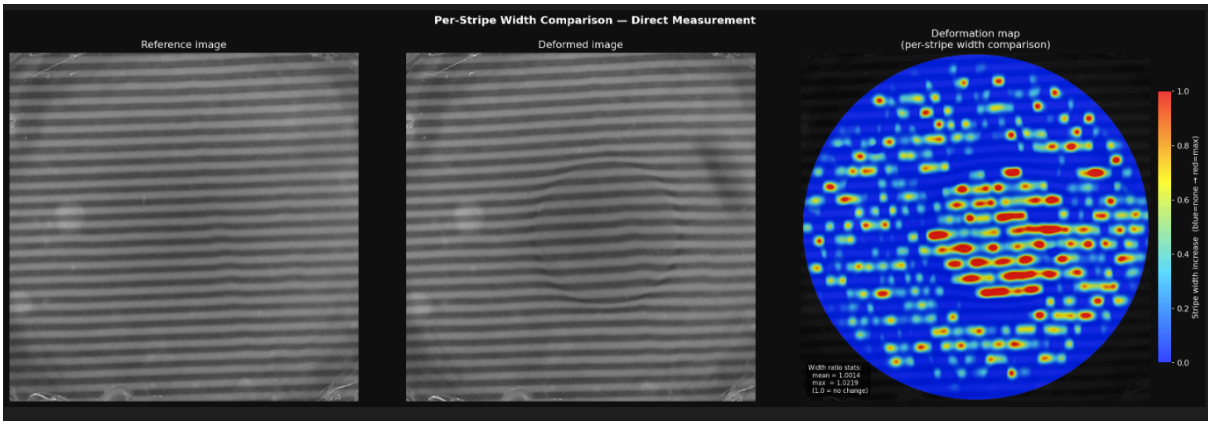
Shape Detection


Elastomer Sensor Diagnostic Panel
Reference
Deformed
Raw Δφ (circular domain, wrapped)
red = pushed right | blue = pushed left
Vertical optical flow fy
red = moved down | blue = moved up
Structure tensor Δθ = θ_def − θ_ref
red = stripes tilt right | blue = tilt left

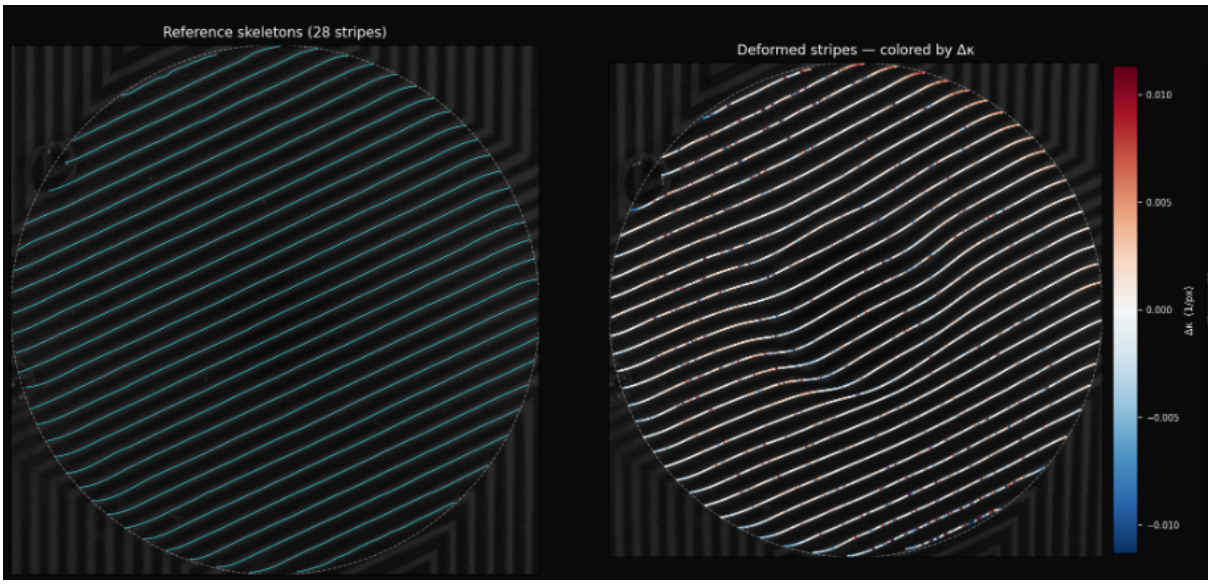

Reference
Deformed
Raw Δφ (circular domain, wrapped)
red = pushed right | blue = pushed left
Vertical optical flow fy
red = moved down | blue = moved up
Structure tensor Δθ = θ_def − θ_ref
red = stripes tilt right | blue = tilt left
2
1
0
−1
−2
20
10
0
−10
−20


Per-Stripe Width Comparison — Direct Measurement
Reference image
Deformed image
Deformation map
(per-stripe width comparison)
Stripe width increase (blue=none → red=max)
1.0
0.8
0.6
0.4
0.2
0.0
Width ratio stats:
mean = 1.0017
max = 1.0244
(1.0 = no change)

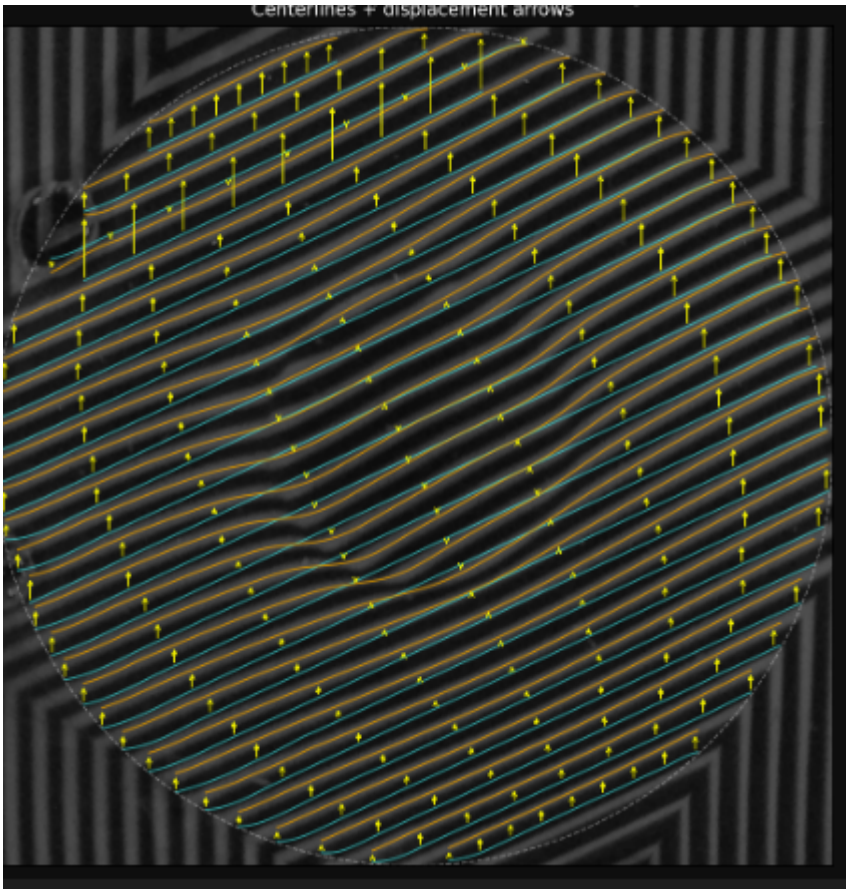

Per-Stripe Width Comparison — Direct Measurement
Reference image
Deformed image
Deformation map
(per-stripe width comparison)
1.0
0.8
0.6
0.4
0.2
0.0
Width ratio stats:
mean = 1.0012
max = 1.0249
(1.0 = no change)

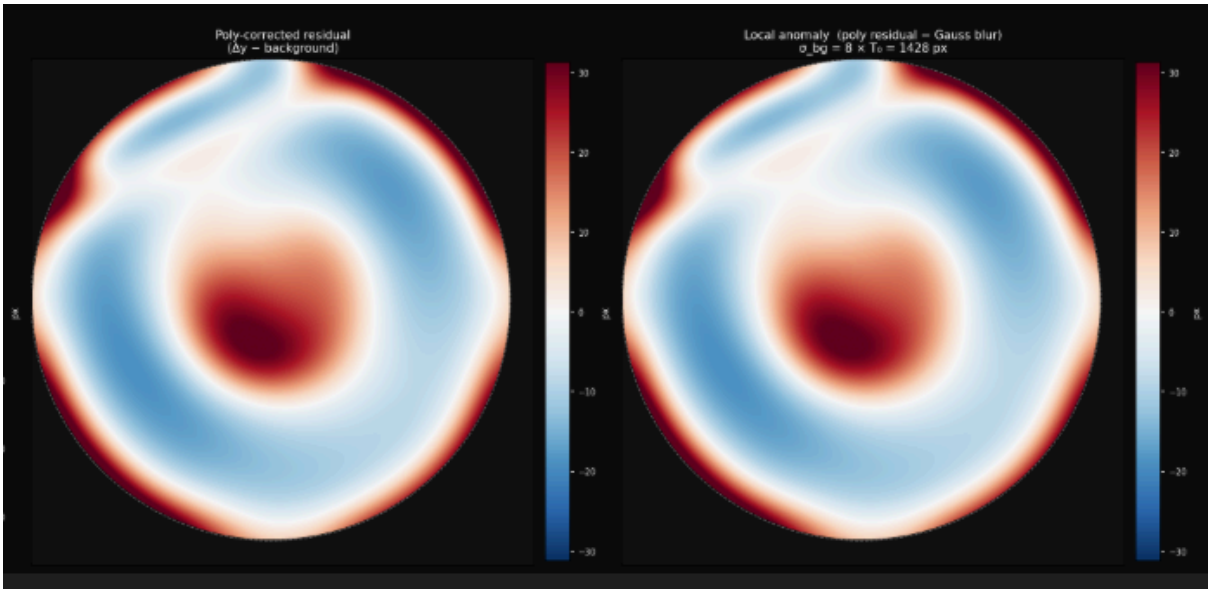

Per-Stripe Width Comparison — Direct Measurement
Reference image
Deformed image
Deformation map
(per-stripe width comparison)
Width ratio stats:
mean = 1.0014
max = 1.0219
(1.0 = no change)
Stripe width increase (blue=none → red=max)
1.0
0.8
0.6
0.4
0.2
0.0

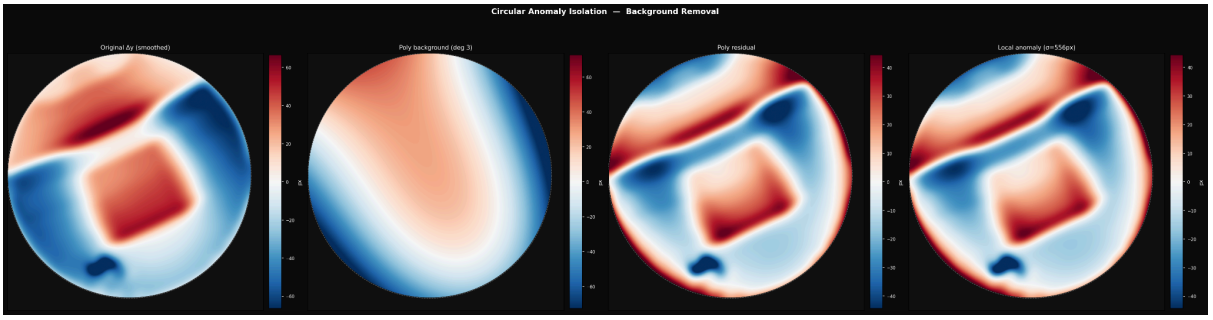
Curvature


Reference skeletons (28 stripes)
Deformed stripes — colored by Δκ
0.010
0.005
0.000
−0.005
−0.010
Δκ (1/px)

Centerline Displacement

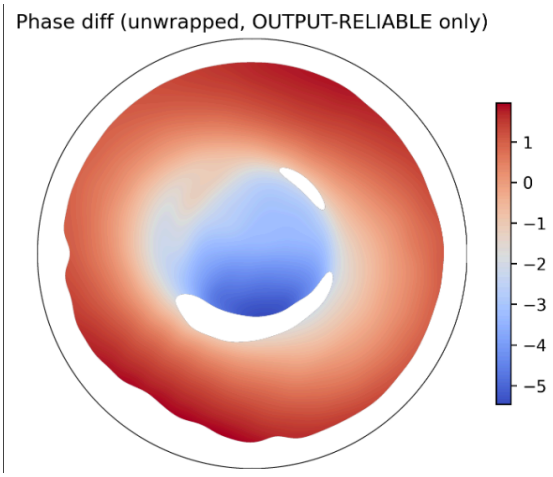


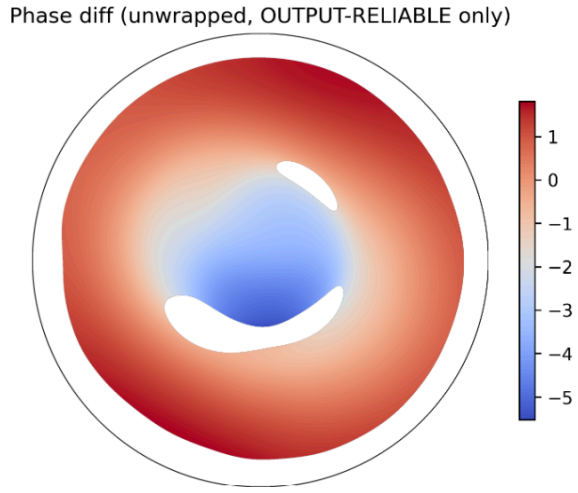




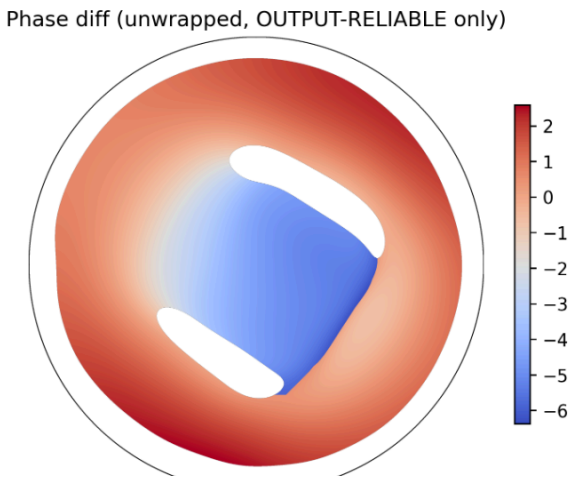
Rim

Flat Circle:

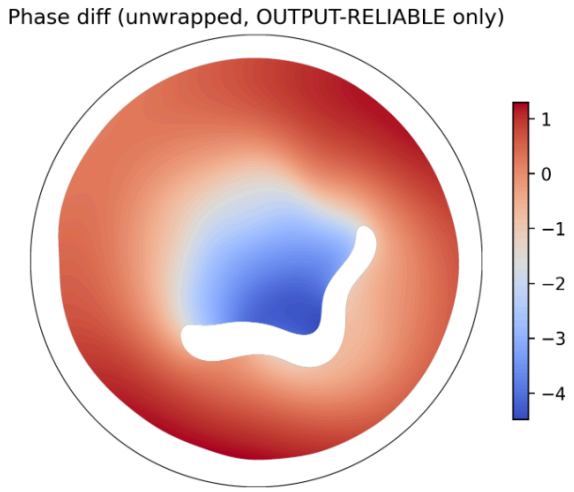


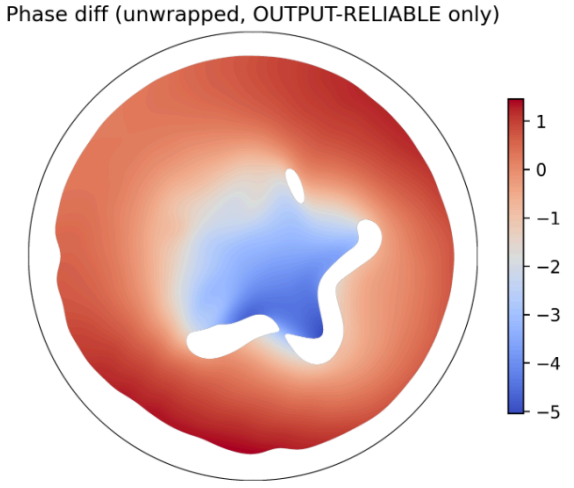


Square:

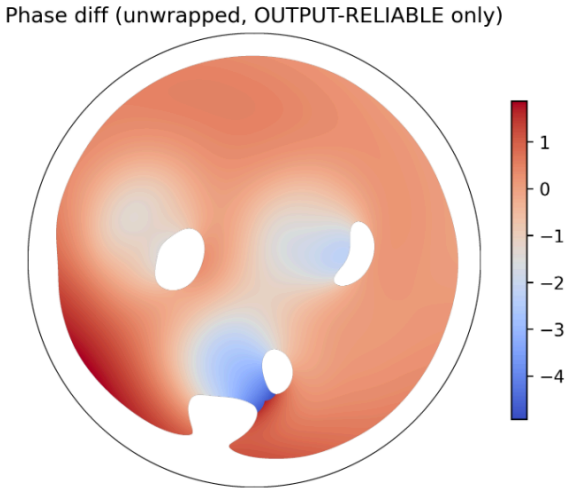

Phase diff (unwrapped, OUTPUT-RELIABLE only)
2
1
0
−1
−2
−3
−4
−5
−6

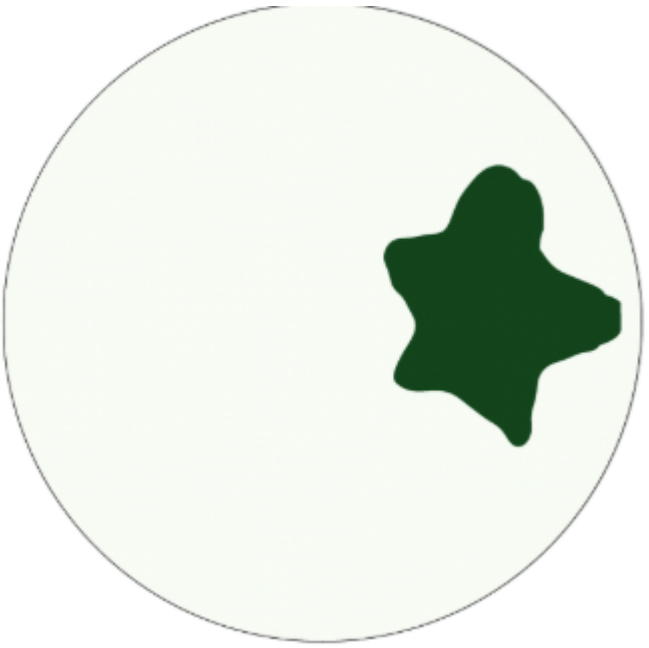
Star:

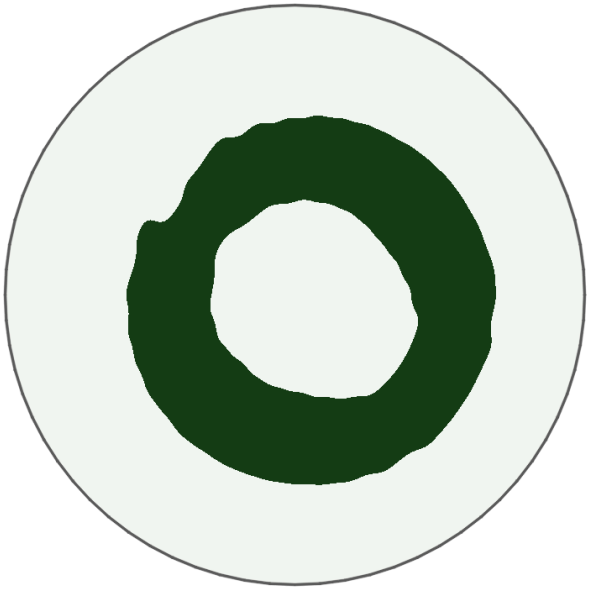

Phase diff (unwrapped, OUTPUT-RELIABLE only)
1
0
−1
−2
−3
−4

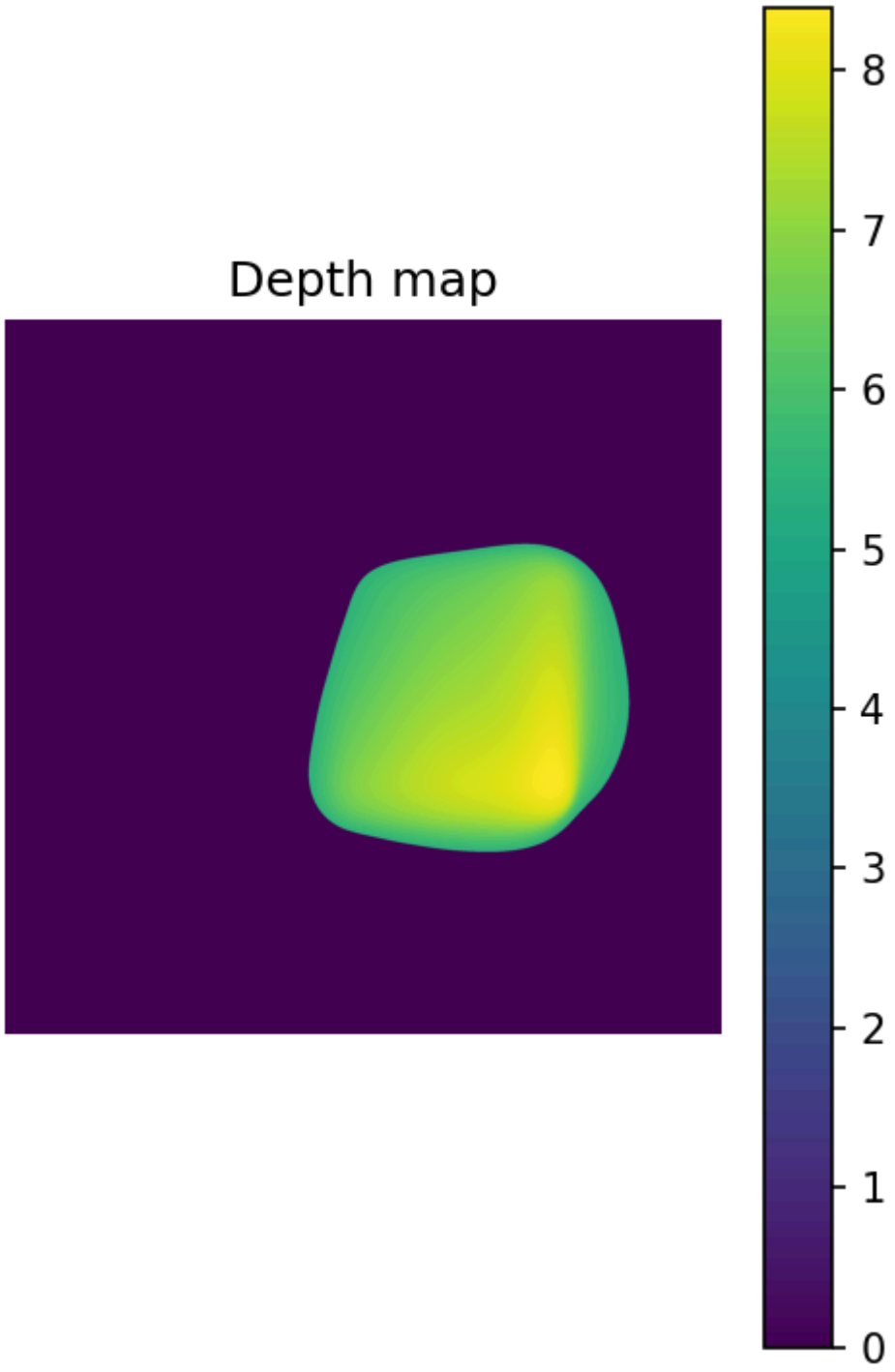

Phase diff (unwrapped, OUTPUT-RELIABLE only)
1
0
−1
−2
−3
−4
−5

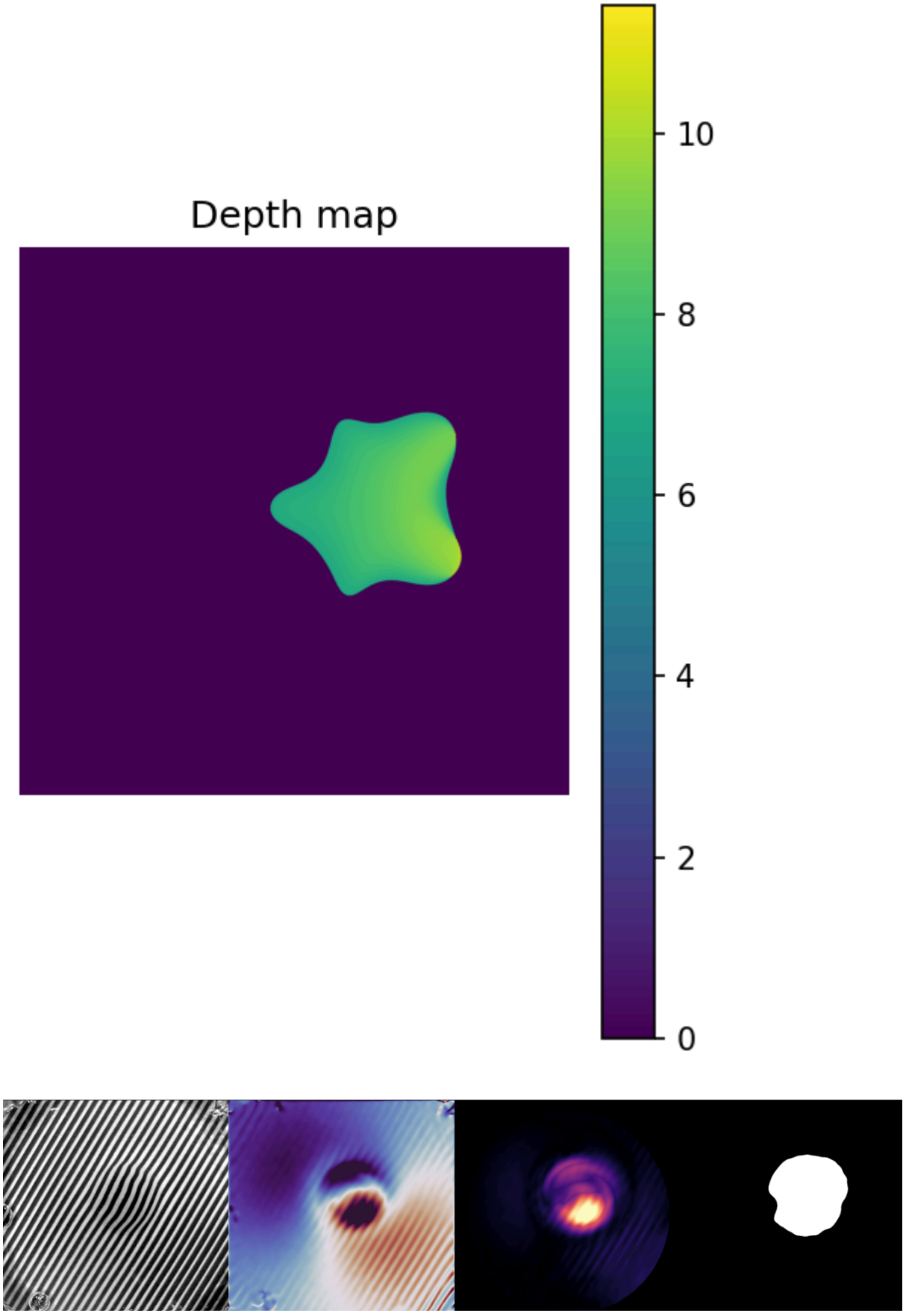
Three circles:





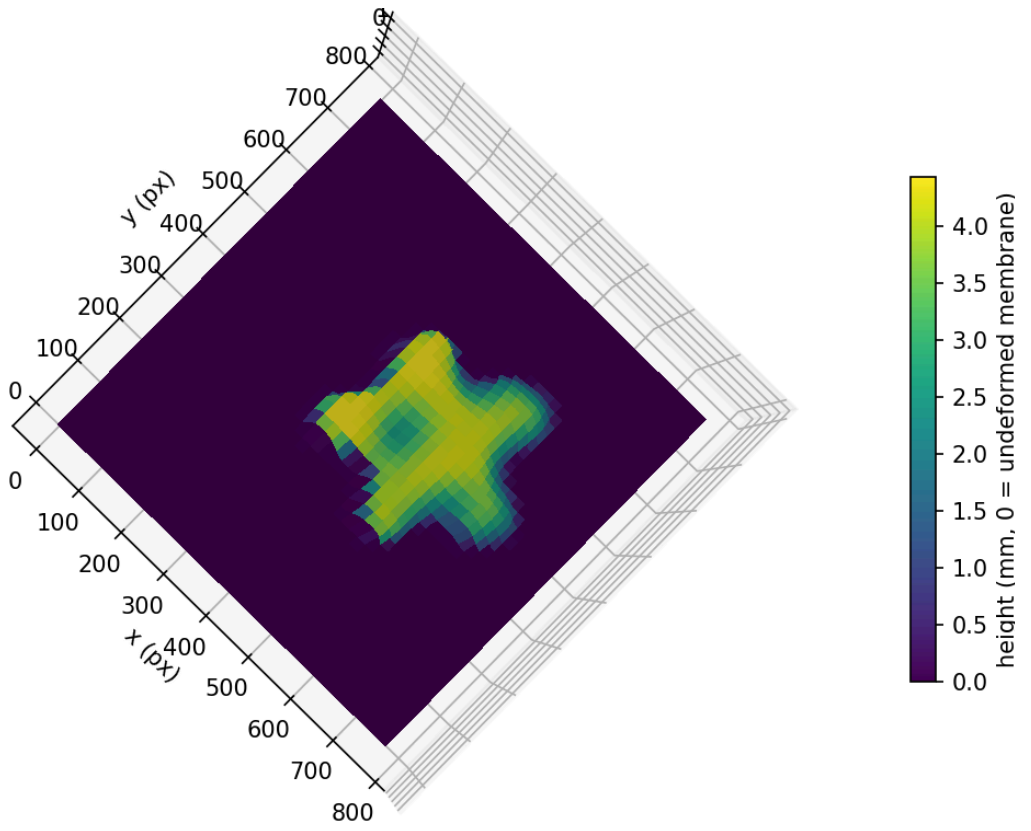



Depth map
8
7
6
5
4
3
2
1
0

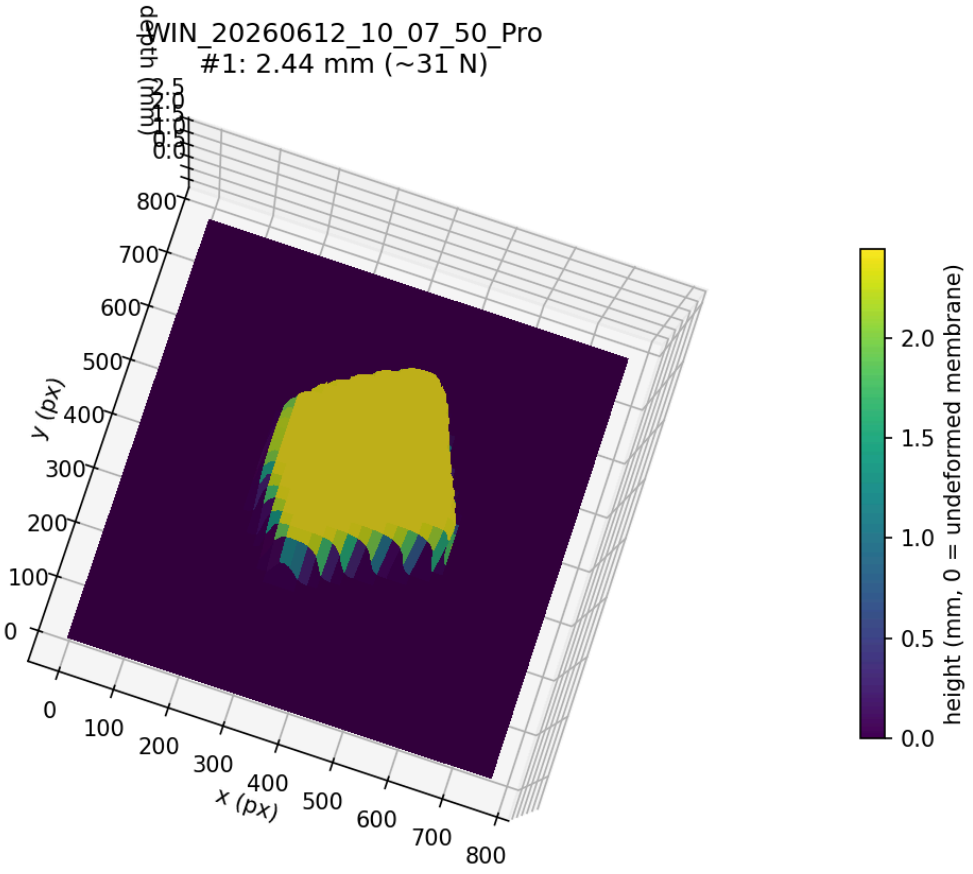

Depth map
10
8
6
4
2
0

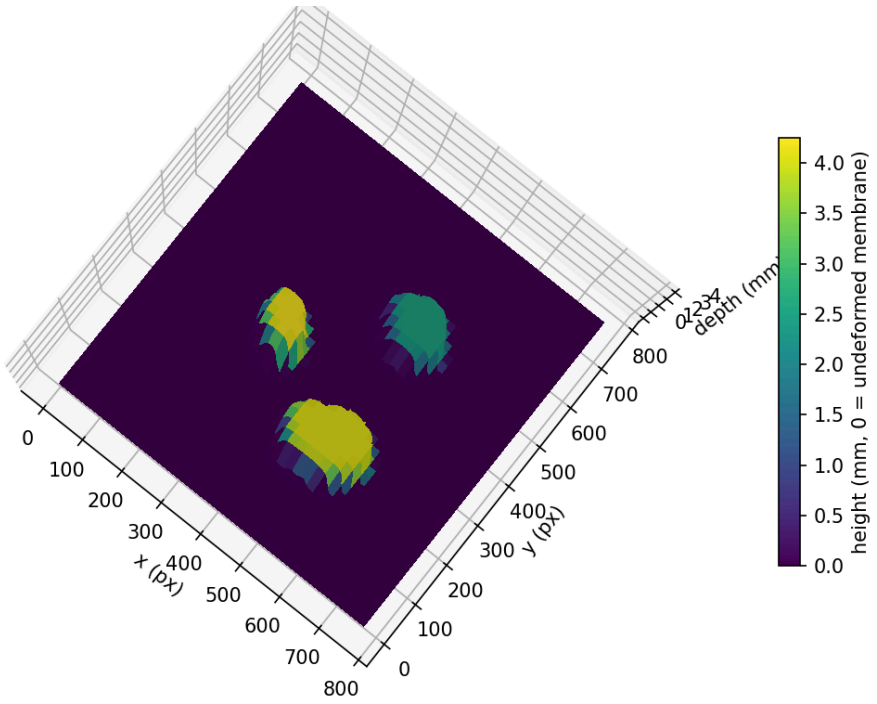
Three Dimensional Reconstruction

Star shape → windowed FFT

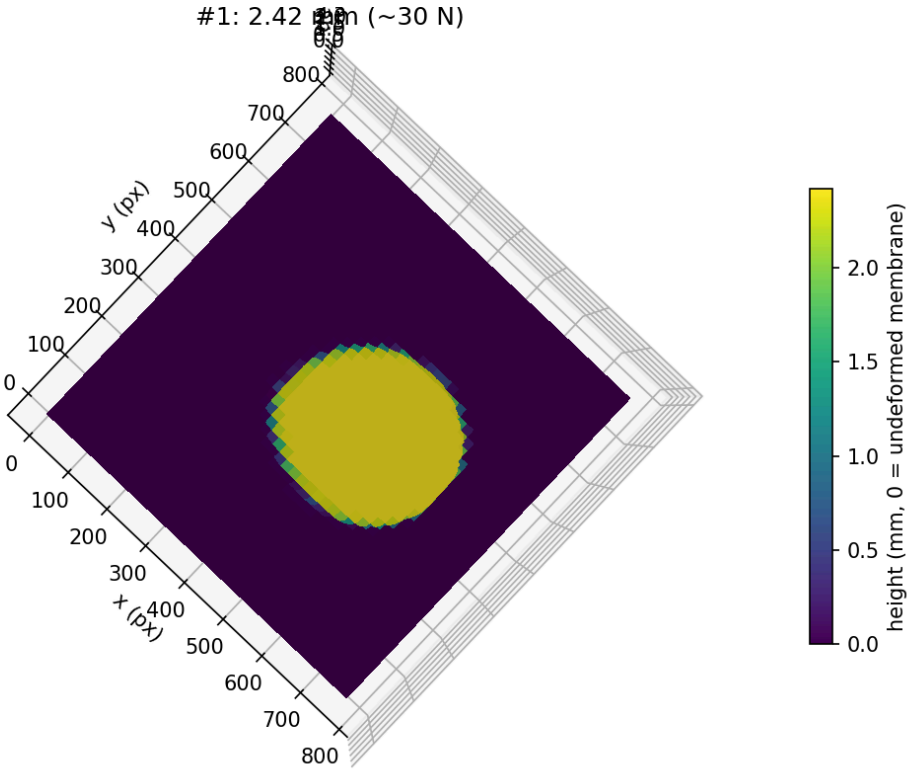


Uniform FFT for square and three circles, but separate shapes can be detected at different heights

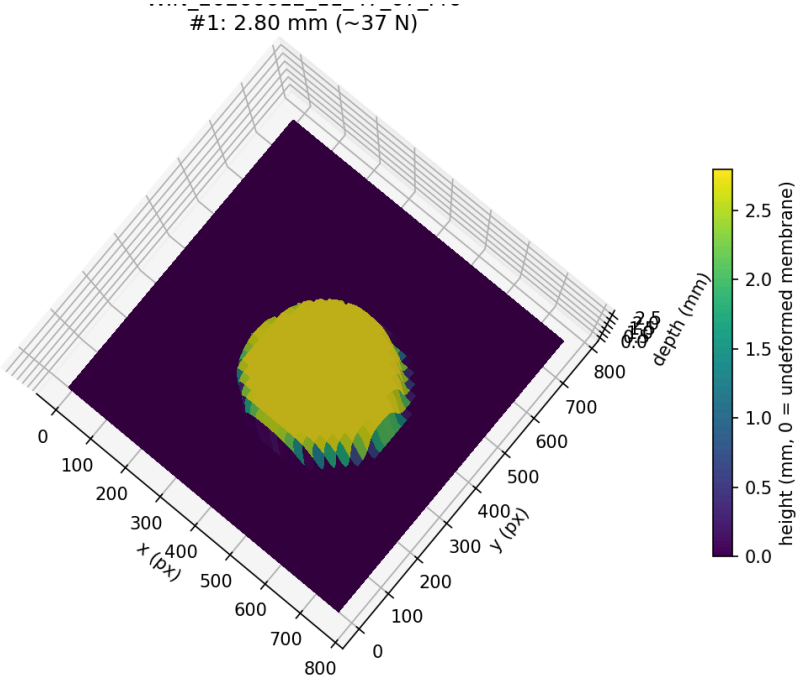




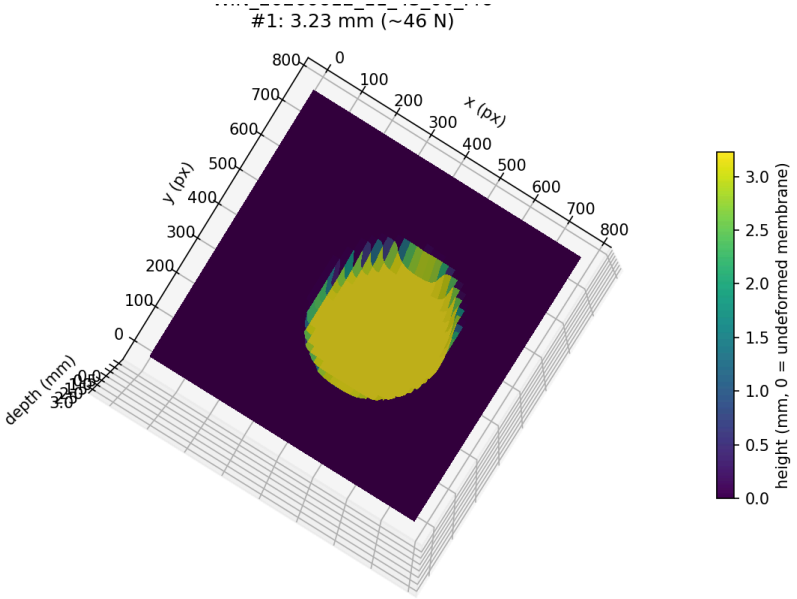
30N applied on circle



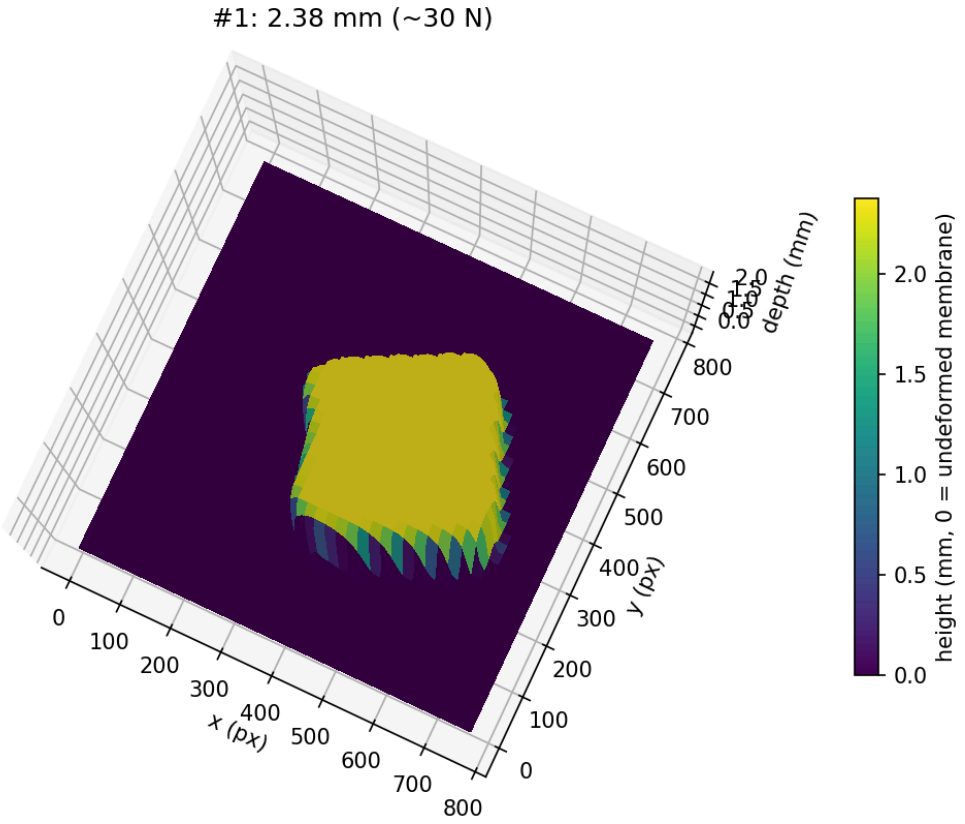
40N applied on Circle



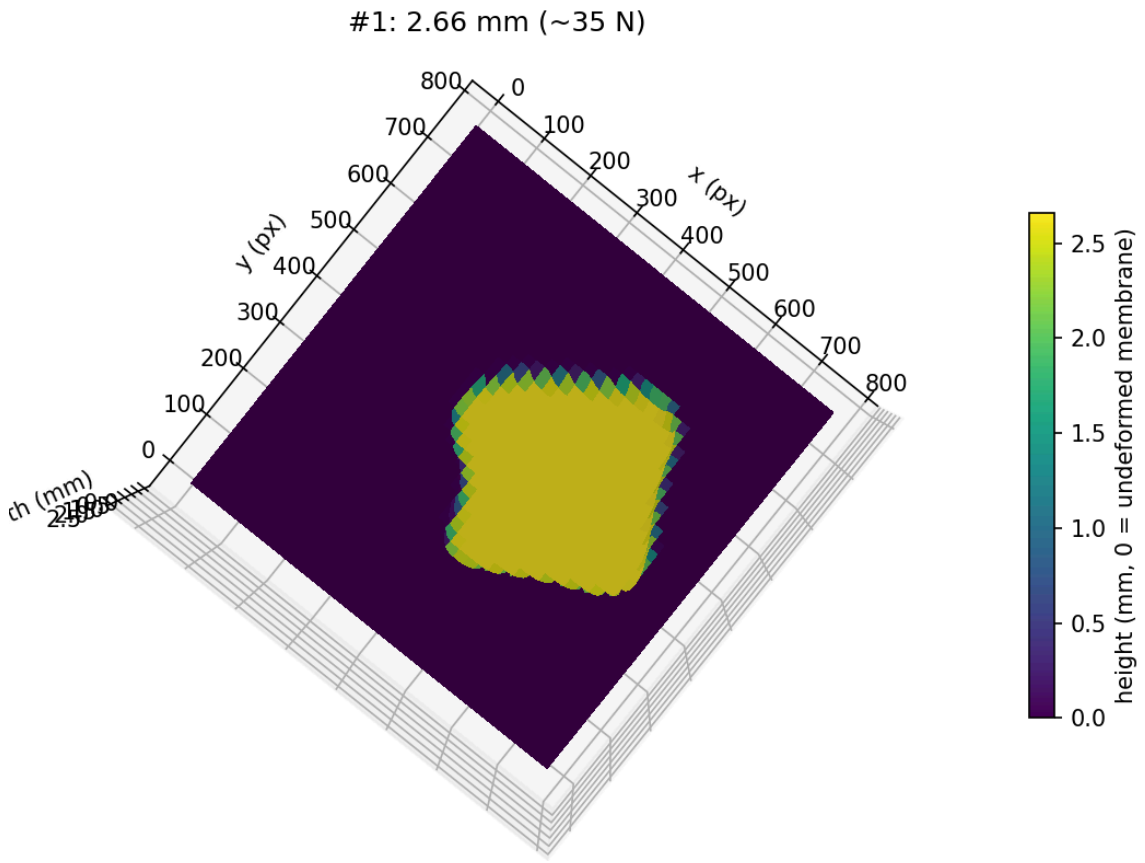
50N applied on circle



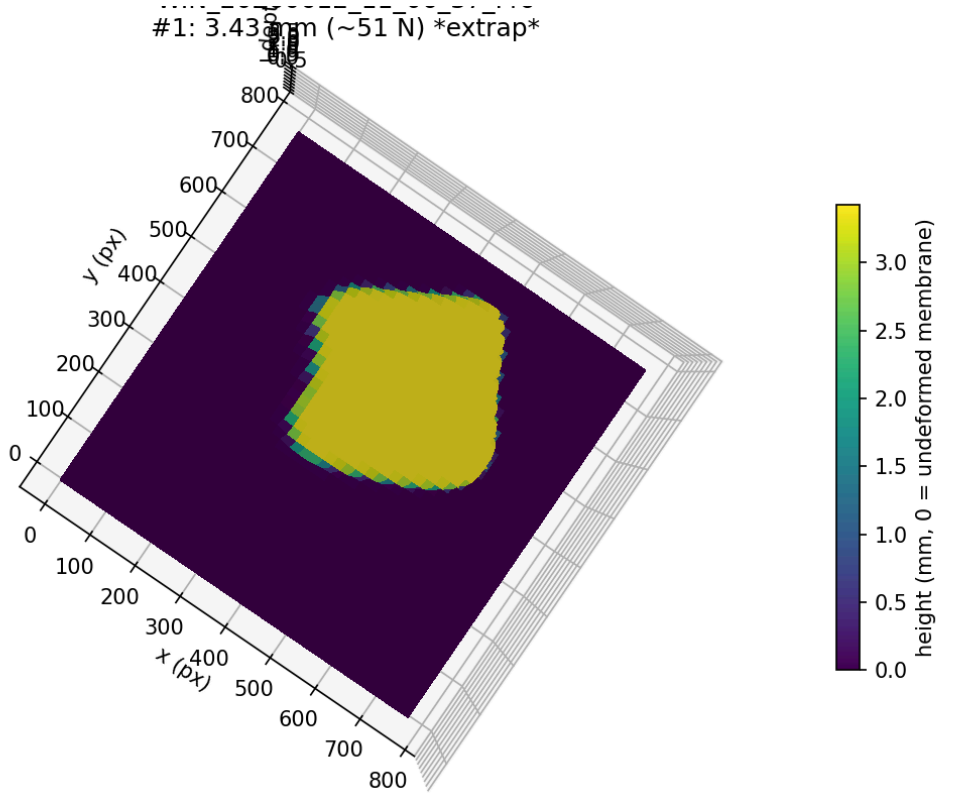
30N applied on square



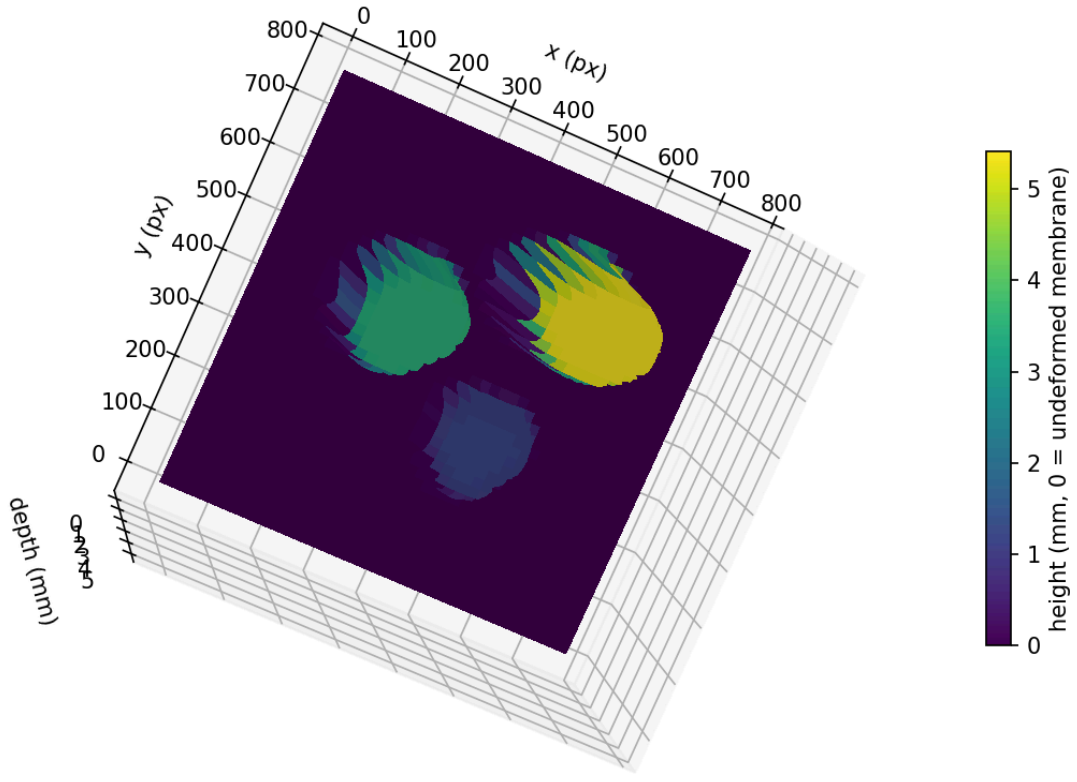
40N applied on square

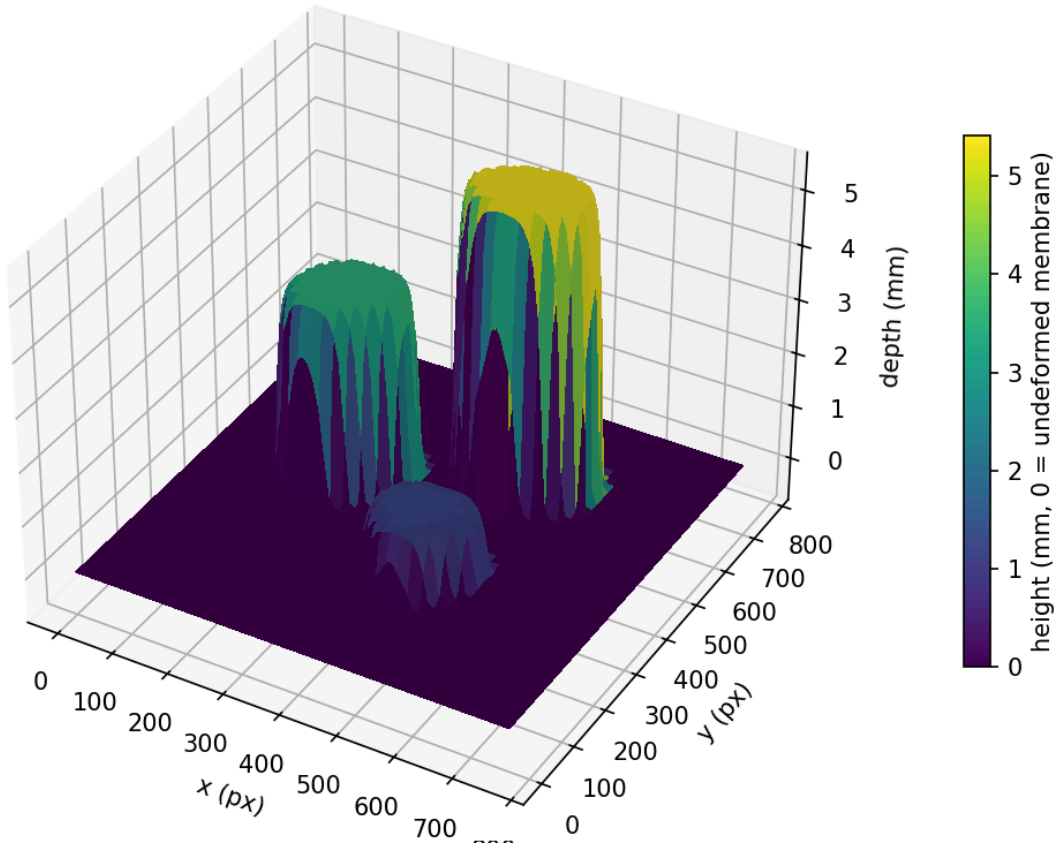


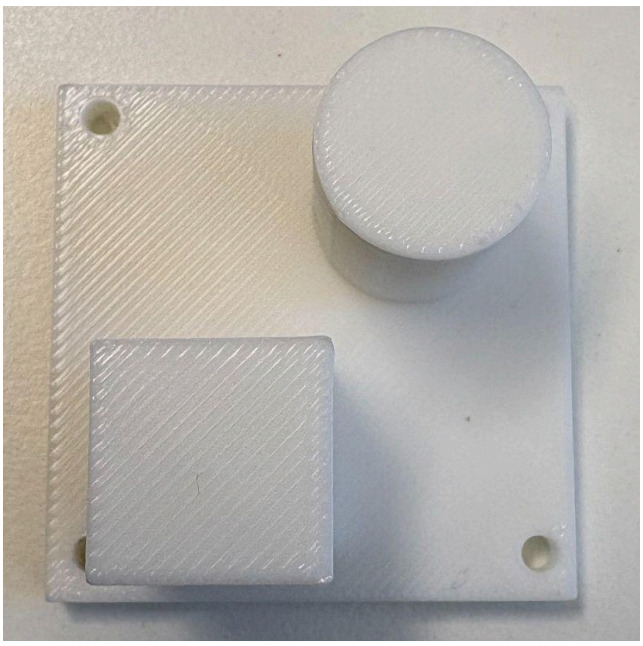
50N applied on square



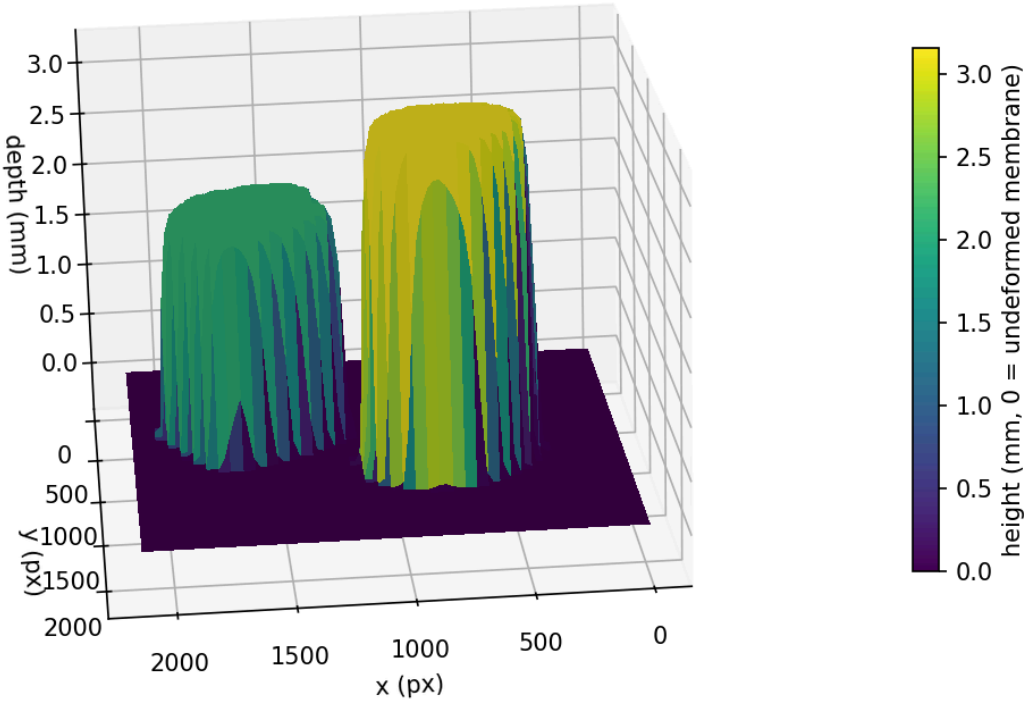
3 circles with different heights

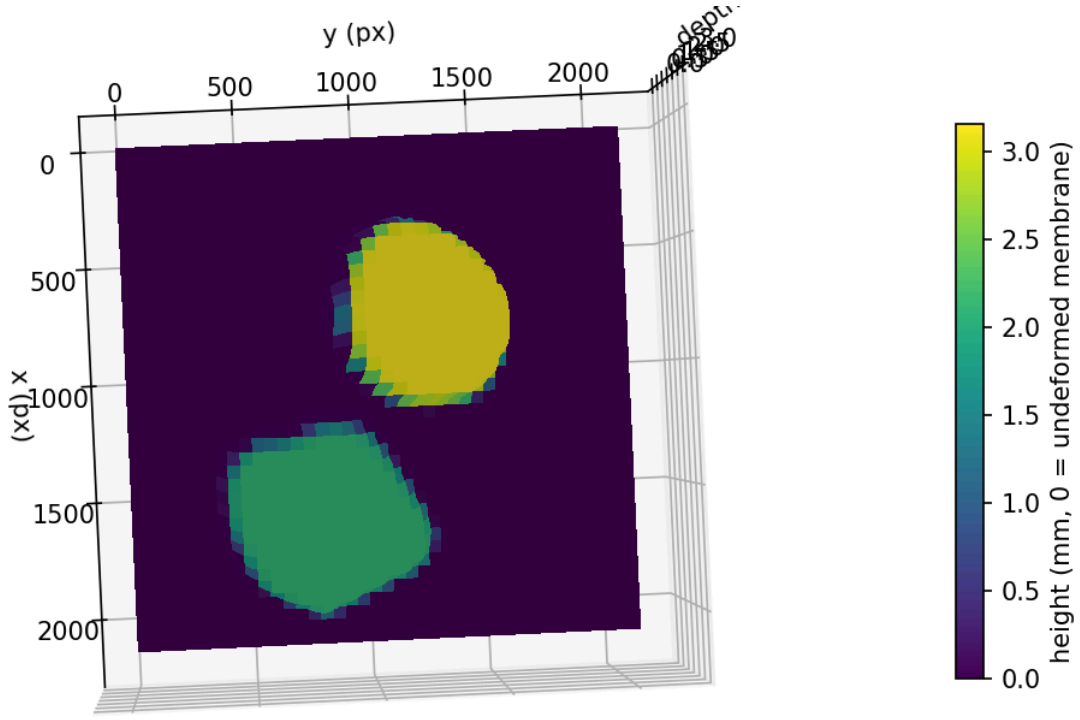



0
100
200
300
400
500
600
700
x (px)
0
100
200
300
400
500
600
700
800
y (px)
0
1
2
3
4
5
depth (mm)
0
1
2
3
4
5
height (mm, 0 = undeformed membrane)



#1: 2.14 mm (~25 N) | #2: 3.16 mm (~45 N)




y (px)
0
500
1000
1500
2000
(xd) x
0
500
1000
1500
2000
height (mm, 0 = undeformed membrane)
3.0
2.5
2.0
1.5
1.0
0.5
0.0